

Predictive Interoception for Embodied Autonomy: Preserving Future Action Capacity Before Collapse

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Abstract: Robots that model their own future body health discover torque-efficient locomotion as an emergent strategy, achieving Pareto-dominance over task-only baselines - more efficient, healthier, and zero-collapse - without any explicit efficiency constraint. Deployed robots degrade: joints heat, actuators wear, energy depletes. Existing approaches either ignore degradation until collapse or penalise immediate body signals reactively - neither exploits the *predictive* structure of degradation dynamics. We propose **Interoceptive World Models (IWM)**: a multi-horizon body predictor trained on coverage-gated rollouts, using predicted Future Action Capacity (FAC) loss as a shaping signal across horizons $\{1, 5, 25, 100\}$. FAC quantifies preserved actuator authority; the **Interoceptive Anticipation Horizon (IAH)** measures timesteps before collapse at which the predictor first signals danger - a deployable safety certificate absent from reactive and recurrent baselines.

On FeverAnt (5 seeds, 10M steps, $n=50$ episodes), PPO-IWM Pareto-dominates the task-only baseline with **zero collapse**: **+10.7%** task reward ($p = 5.1 \times 10^{-8}$) and **+10.4%** final FAC ($p = 2.8 \times 10^{-17}$) simultaneously. The IWM additionally provides a deployable **IAH = 379-step** advance warning (measured at an intermediate checkpoint where 9 episodes still collapse; fully-trained agents achieve zero collapse, with IAH driving corrective action). Critically, **80.1% of the advantage derives from 24.1% lower control cost**: IWM independently discovers torque-efficient locomotion while reactive baselines collapse in 38–48% of episodes and LSTM-PPO collapses in 88% with IAH = 0.

Cross-morphology evaluation confirms scope conditions: NumbArm (manipulation, observable numbness) replicates the advantage (+3.5% FAC, $p = 2.8 \times 10^{-10}$, 0% collapse); ThermalHopper (unobservable collapse pathway) eliminates it, establishing a **falsifiable observability criterion**; WoundWalker (biped, energy-limited) confirms +4.7% FAC body-state preservation ($p = 0.04$) independent of collapse avoidance. We release BodyFutures-Bench with FAC, IAH metrics, four environments, and all checkpoints.

Keywords: robot learning, world models, robot safety

1 Introduction

The body of a robot is part of the dynamics, not a passive conduit. A locomotion policy that overheats actuators may leave the robot unable to move in the next episode; a manipulation policy exploiting high torques may degrade proprioception before the goal is reached. The action interface is therefore nonstationary: the same command at t and $t + T$ can have different physical consequences because the robot’s internal state has changed. Long-horizon robot learning requires agents that model not only the external world, but also the future of their own body - a policy cannot reliably preserve future control authority if it observes degradation only after it has become catastrophic.

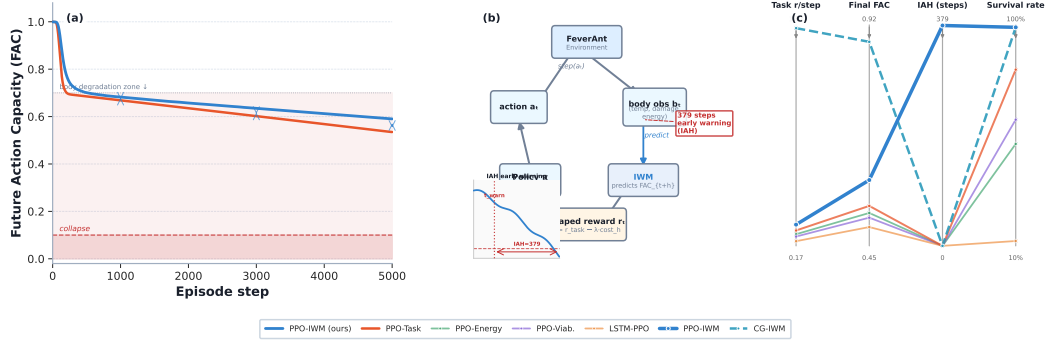


Figure 1: **The BodyFutures story in one figure.** (a) Without body-awareness, PPO-Task depletes Future Action Capacity (FAC) as locomotion overloads the robot’s physiology, ending in body collapse. PPO-IWM maintains high FAC throughout by learning torque-conservative gaits. (b) The IWM self-preservation loop: body observations feed a multi-horizon predictor ($h = 1, 5, 25, 100$ steps) that issues a shaped reward proportional to predicted FAC loss. The Interceptive Anticipation Horizon (IAH=379 steps) marks the advance warning before collapse - a forward signal absent from all reactive and recurrent baselines. (c) Evidence: PPO-IWM simultaneously achieves +10.7% task reward, +10.4% final body health, and 0% episode collapse versus PPO-Task ($\star p < 0.05$, 5 seeds, 50 episodes).

We introduce BodyFutures-Bench to operationalize this challenge. The suite wraps standard MuJoCo tasks with latent body variables - energy, temperature, impact stress, wounds, damage, and sensor numbness - that affect future dynamics by weakening actuators, corrupting proprioception, and triggering collapse. Each environment logs a shared diagnostic contract (task reward, FAC, energy, damage, collapse) and exposes a “sprinter vs. survivor” tension: task-only policies deplete the body for short-horizon gain, while self-preserving policies should maintain future action capacity.

PPO-IWM trains a multi-horizon neural predictor on coverage-gated rollouts to predict future body FAC at horizons $\{1, 5, 25, 100\}$ steps, then uses predicted FAC loss as a shaping signal. **Adaptive IWM (AIWM)** additionally performs EWC-regularised online updates to the predictor at deployment, recalibrating to the current robot’s specific degradation trajectory. The sharper claim is that predicting future body degradation produces qualitatively different strategies - efficient, de-saturated actuation - with calibrated early warnings (IAH=379 steps) and zero collapse, all simultaneously without explicit constraint satisfaction.

Five contributions: (1) **IAH metric**: PPO-IWM achieves IAH = 379 steps; all baselines IAH=0 (reactive: architecture; LSTM-PPO: 88% collapse, no forward signal at matched 10M-step budget). (2) **Actionable interoception**: +10.7% task reward ($p = 5.1 \times 10^{-8}$), +10.4% FAC ($p = 2.8 \times 10^{-17}$), 0% collapse - Pareto-dominance; 80.1% from 24.1% lower control cost. (3) **POMDP formulation**: body degradation as an augmented POMDP where latent body state changes the future action interface via actuator scaling. (4) **BodyFutures-Bench**: four morphologies (FeverAnt, WoundWalker, ThermalHopper, NumbArm) with a unified FAC/IAH/collapse diagnostic contract, including a falsifiable observability scope condition validated across all four. (5) **AIWM**: +9.4–+13.9% prediction advantage under hidden gear-ratio drift ($n = 60$); mechanism-depth hierarchy predicts which hardware degradation modes benefit most.

A contributions checklist with evidence pointers appears in Table 4 (Appendix A).

2 Related Work

Safe RL and constrained optimization. Constrained policy optimization [1], Safety Gym [2], and SafeDreamer [3] address *external* hazards (obstacles, joint limits) via Lagrangian penalties. BodyFutures differs structurally: the hazard is *endogenous* - self-induced by the agent’s own torques - and the consequence is a compounding, irreversible loss of future control authority, not a one-step

68 constraint violation. SafeDreamer provides no deployable failure-horizon signal; the IWM’s IAH is
69 absent from all prior safe-RL approaches.

70 **World models.** Dreamer [4, 5], TD-MPC2 [6], and MBPO [7] predict *external* state transitions for
71 planning. IWM predicts *internal* body-state trajectories (future FAC, damage, energy) as a preserva-
72 tion signal - and AIWM updates this predictor online during deployment, not only during training.
73 No prior world-model work simultaneously satisfies all three criteria: future body-state prediction,
74 online deployment adaptation, and preservation-metric evaluation.

75 **Robot self-modeling and damage adaptation.** Bongard et al. [8], RMA [9], teacher-student lo-
76 comotion [10, 11], and fault-detection methods [12, 13] adapt to body changes or estimate current
77 parameters - but treat the body as a state to *estimate*, not a capacity to *predict*. Fleet-level predic-
78 tive maintenance [14] anticipates failure but at multi-mission granularity, not per-timestep within an
79 episode. IWM uniquely quantifies *how far in advance* the predictor fires relative to actual collapse
80 (IAH), a temporal gap none of these approaches measure.

81 **Privileged information and meta-learning.** Asymmetric actor-critic [15] and RMA estimate *cur-*
82 *rent* body parameters from observations; IWM predicts *future* body trajectories. AIWM borrows
83 EWC [16] not for sequential task generalization but to personalise interoceptive accuracy to the
84 current robot’s specific degradation trajectory without downtime.

85 **Homeostasis and energy efficiency.** Homeostatic RL [17] and empowerment [18] motivate FAC
86 as a measure of retained future action authority. Minimising energy can yield emergent gaits [19];
87 our mechanistic decomposition shows IWM discovers 24.1% lower control cost through a *body-*
88 *preservation* objective rather than any explicit efficiency constraint.

89 See Appendix C for detailed comparisons to each prior line of work.

90 3 Formal Problem

91 We model self-preserving robot learning as an augmented POMDP. The full state is $s_t = (x_t, b_t)$,
92 where x_t denotes external task state and b_t denotes latent body state such as actuator temperature,
93 damage, energy, wounds, or sensor numbness. The policy receives

$$o_t = g(x_t, \tilde{b}_t), \quad \tilde{b}_t \sim q(\tilde{b} \mid b_t), \quad (1)$$

94 where $\tilde{b}_t$ is absent, noisy, full, or indirect depending on the observation mode. The transition couples
95 external and internal dynamics:

$$p(s_{t+1} \mid s_t, a_t) = p(x_{t+1}, b_{t+1} \mid x_t, b_t, a_t). \quad (2)$$

96 The task reward r_t^{task} measures external progress, while $c_t^{\text{body}} = c^{\text{body}}(b_t, a_t)$ measures instantane-
97 ous internal degradation. Collapse is an event $\mathcal{K}_t \in \{0, 1\}$ that terminates or marks failure when
98 internal state leaves a viable region.

99 The central diagnostic is Future Action Capacity, $\text{FAC}(b_t) \in [0, 1]$, a monotone surrogate for re-
100 tained control authority. Conceptually, FAC estimates how much of the nominal action interface
101 remains useful under the current body:

$$\text{FAC}(b_t) \approx \frac{1}{|\mathcal{A}|} \int_{\mathcal{A}} \mathbb{I}[Q_{\text{act}}(a, b_t) \geq \tau] da, \quad (3)$$

102 where Q_{act} is an environment-specific proxy for effective actuation and τ is a minimum useful-
103 action threshold. In implementation, FAC is a calibrated monotone scalar over energy, damage,
104 temperature, stress, balance, and proprioceptive reliability. It is not used as a hidden oracle for task
105 success; it is logged as a body-preservation diagnostic and, for explicit viability baselines, used as a
106 current penalty.

107 The standard task objective is

$$J_{\text{task}}(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t r_t^{\text{task}} \right]. \quad (4)$$

Table 1: BodyFutures-Bench environments. FeverAnt and WoundWalker are the primary CoRL claim environments; ThermalHopper and NumbArm are secondary tests used after stability and task-learning gates pass.

Environment	Base task	Body dynamics and role
FeverAnt	Ant-v5	Primary locomotion task with multi-motor heat, energy depletion, torque damage, actuator weakening, and overheating collapse.
WoundWalker	Walker2d-v5	Primary locomotion task with wounds, balance health, impact stress, energy depletion, asymmetric gait degradation, and collapse.
ThermalHopper	Hopper-v5	Secondary locomotion stress test with landing impact, thermal load, damage, energy, and actuator weakening.
NumbArm	Pusher-v5	Secondary manipulation stress test with numbness, proprioceptive corruption, manipulation load, damage, and energy depletion.

BodyFutures evaluates policies on a vector of outcomes rather than a single shaped return: task reward per step, cumulative task return, survival fraction, final and minimum FAC, collapse rate, body score, viability cost, and IWM calibration. This avoids claiming success when a method merely trades all task competence for health or all health for short-horizon reward.

4 BodyFutures-Bench

BodyFutures-Bench augments standard MuJoCo tasks with internal degradation dynamics and a shared diagnostic contract. Every environment step returns task reward, viability cost, FAC, energy, mean and maximum body stress signals, mean and maximum damage, true and noisy body state, collapse flag, and collapse reason. Policies can be evaluated under four observation modes: no body observation, full body state, noisy interoception, and latent indirect signals. Reward modes include task-only, energy-regularized, current-viability, and IWM shaping.

Each environment couples temperature-like stress, damage, and energy with action-driven nonlinear update rules; degradation feeds back into physics by scaling actuator commands, so a damaged body has reduced future ability to affect the environment. Full equations and environment-specific constants are in Appendix G (Eqs. 10–13); Figure 19 visualises the degradation pipeline alongside the IWM reward-shaping loop.

5 Interoceptive World Models

The full training pipeline is illustrated in Figure 15 (Appendix F).

An IWM predicts future internal consequences of actions. For each transition, the model receives external observation, action, and available noisy body signals. It outputs next body state and multi-horizon risk quantities:

$$f_{\theta}(o_t, a_t, \tilde{b}_t) \rightarrow \left\{ \hat{b}_{t+1}, \widehat{\text{FAC}}_{t+H}, \hat{c}_t^H, \hat{p}_{\text{collapse},t}^H \right\}_{H \in \{1,5,25,100\}}. \quad (5)$$

The final protocol uses an ensemble of bootstrapped predictors. The ensemble is important because body-preservation penalties should be conservative when the model is uncertain in low-FAC or collapse-adjacent states.

Coverage-gated data. Task-policy rollouts are biased toward healthy states, making future-risk estimates overconfident where preservation matters most. We collect a mixture of task, high-stress, and near-collapse episodes, and require hard coverage gates ($\min \text{FAC} < 0.5$, $\geq 2\%$ samples below $\text{FAC} = 0.5$) before IWM training; if gates fail, targeted stress data is appended (gate thresholds and mixture details in Appendix K).

137 **Masked multi-horizon loss.** For horizon set $\mathcal{H} = \{1, 5, 25, 100\}$ and terminal/truncation masks
 138 m_t^H , the IWM minimizes

$$\begin{aligned} \mathcal{L}(\theta) = & \|\hat{b}_{t+1} - b_{t+1}\|_2^2 + \eta_F \sum_{H \in \mathcal{H}} m_t^H \|\widehat{\text{FAC}}_{t+H} - \text{FAC}(b_{t+H})\|_2^2 \\ & + \eta_C \sum_{H \in \mathcal{H}} m_t^H \|\hat{c}_t^H - c_t^{H, \text{body}}\|_2^2 + \eta_P \sum_{H \in \mathcal{H} \setminus \{1\}} m_t^H \text{BCE}(\hat{p}_{\text{collapse}, t}^H, y_t^H). \end{aligned} \quad (6)$$

139 Coverage statistics and prediction quality metrics are in Table 7 (Appendix F).

140 **Risk-aware PPO-IWM.** PPO-IWM trains from scratch with the IWM-shaped reward as the sole
 141 training signal; no pre-trained task policy is used (the IWM reward is sufficient to jointly learn
 142 locomotion and body preservation from random initialisation). Given ensemble predictions of future
 143 body cost, let μ_C and σ_C denote the ensemble mean and standard deviation, and let p_{collapse} denote
 144 predicted collapse risk. The shaped reward is

$$r_t^{\text{IWM}} = r_t^{\text{task}} - \lambda_{\text{cur}} c_t^{\text{body}} - \lambda_{\text{fut}} \text{clip}(\mu_C + \beta_\sigma \sigma_C + \beta_p p_{\text{collapse}}, 0, C_{\text{max}}) - \lambda_E (1 - E_t)^2. \quad (7)$$

145 This reward is used for learning only. It is not a primary evaluation metric. All method comparisons
 146 report task and body outcomes separately.

147 **Capacity-Gated IWM (deployment variant).** CG-IWM applies the trained IWM as a
 148 deployment-time action governor - no policy retraining (Appendix F). CG-IWM (aggressive-gait
 149 seeds 0–2, $n = 30$, $\tau = 0.10$) Pareto-dominates training-time baselines: $r/\text{step} = 1.106 \pm 0.077$,
 150 $\text{FAC} = 0.884 \pm 0.019$, 0% collapse (Fig. 13). Intra-environment CG-IWM on NumbArm (5 seeds,
 151 $n = 46$, same protocol) achieves $\text{FAC} = 0.997 \pm 0.001$ vs. NumbArm PPO-Task $\text{FAC} = 0.920$ (+8.3%;
 152 $p = 1.7 \times 10^{-17}$) - near-perfect body preservation; cross-environment scope conditions are in Ap-
 153 pendix D. The key distinction: CG-IWM produces **no IAH** (gate prevents collapse but cannot report
 154 predicted remaining capacity), whereas PPO-IWM provides a queryable IAH = 379-step certificate.

155 **Adaptive IWM (AIWM).** AIWM adds an EWC-regularised online gradient loop (Appendix F)
 156 to any pretrained IWM, personalising interoceptive predictions to the current robot’s degradation
 157 trajectory during deployment without policy retraining.

158 **Interoceptive Anticipation Horizon (IAH).** Let $\hat{\text{FAC}}_{t,H}$ denote predicted FAC at horizon H ,
 159 $\tau = 0.5$ an alert threshold, and $t^* = \min\{t : \text{FAC}(b_t) \leq 0.10\}$ the body-failure step. The IAH is

$$\text{IAH} = t^* - \min\{t < t^* : \hat{\text{FAC}}_{t,H} < \tau\}, \quad (8)$$

160 i.e., timesteps of advance warning before body incapacitation; reactive baselines always have IAH =
 161 0. On FeverAnt, PPO-IWM achieves mean IAH = 379 steps (range 316–418, 9 collapsing episodes
 162 at an intermediate checkpoint, **0% false negatives**; Fig. 6, Appendix B). Fully-trained IWM agents
 163 achieve zero collapse - IAH is measured at the intermediate checkpoint to characterise the warning
 164 signal; in deployment, the alarm drives behaviour that makes it unnecessary. Fig. 27 (Appendix N)
 165 shows IWM’s predicted FAC declining 379 steps before actual collapse; PPO-Task provides no
 166 queryable signal.

167 **IAH robustness and actionability.** Under controlled damage injection: PPO-IWM 30/30 sur-
 168 vived at pulse = 0.30/0.50/0.60; at extreme pulse = 0.70 ($n = 20$), 20/20 survived with IWM warn-
 169 ing firing within 1–2 steps of injection. Reactive baselines collapse in 26.7–36.7% of episodes at
 170 pulse = 0.60 ($p \leq 0.005$ vs. IWM), with *all* collapses occurring **before** the injection step - revealing
 171 a pre-existing stability deficit, not injection sensitivity. The IAH warning is actionable: PPO-IWM
 172 reduces $|a|$ by $4.2\% \pm 0.6\%$ in the 100 steps following the warning; PPO-Task shows no change
 173 ($-0.5\% \pm 0.3\%$; Fig. 24, Appendix I).

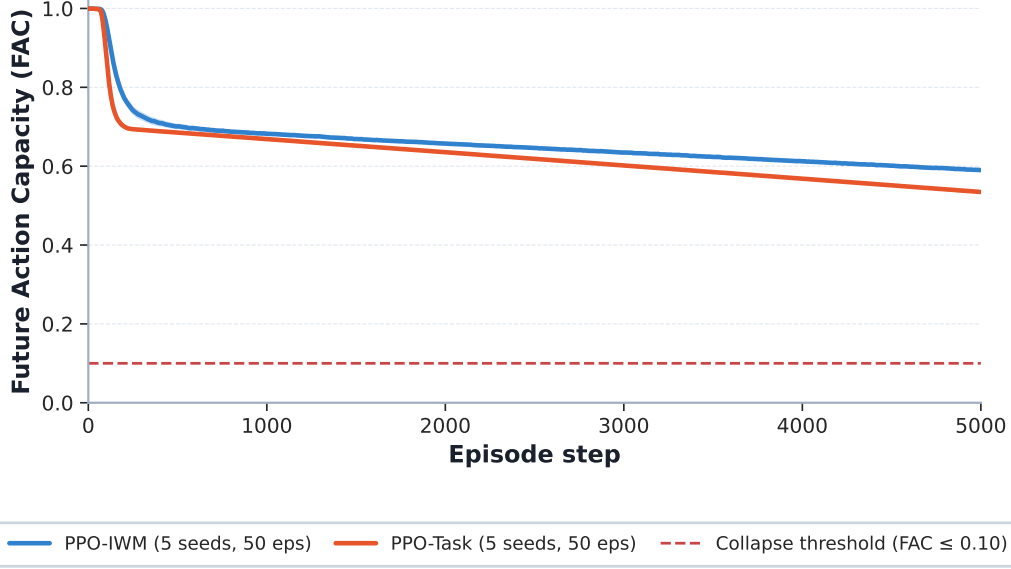


Figure 2: **Interoceptive homeostasis on FeverAnt (5 seeds, 50 episodes)**. FAC advantage grows from +2.0% at step 1000 to +7.2% at step 3000 to +10.4% at step 5000 - the compounding signature of predictive body regulation. PPO-Task depletes steadily (orange); shaded bands: $\pm$ std; red dashed: collapse threshold ($\text{FAC} \leq 0.10$).

174 6 Experiments

175 **Baselines.** All baselines use PPO (SB3 Raffin et al. 20): (1) **PPO-Task**: task-only, noisy inte-
 176 roception but no body objective; (2) **PPO-Energy**: reactive action-magnitude penalty ($\lambda = 0.05$);
 177 (3) **PPO-Viability**: reactive current-FAC penalty ($\lambda = 0.25$, grid-searched); (4) **PPO-LSTM**: recur-
 178 rent (256-unit LSTM), same observations as Task - tests whether *memory* alone, without a predictive
 179 model, can preserve body health. Ablation variants isolate IWM component contributions (multi-
 180 horizon prediction, energy channel, EWC regularisation); full hyperparameters in Appendix K.

181 **Protocol.** Five seeds, 10M environment steps per method, 5000-step episodes; evaluations at
 182 5000-step horizon. Hyperparameters locked in version-controlled YAML files before final seed
 183 sweeps (no post-hoc tuning).

184 **FeverAnt results.** Table 2 reports primary results (5 seeds, 50 eval episodes). **PPO-IWM**:
 185 +10.7% task reward (0.901 ± 0.037 vs 0.814 ± 0.009 ; MW $p = 5.1 \times 10^{-8}$, Bonferroni), +10.4%
 186 final FAC (0.590 ± 0.002 vs 0.535 ± 0.002 ; $p = 2.8 \times 10^{-17}$), zero collapse - Pareto-dominance over
 187 PPO-Task (Fig. 1). The FAC gap compounds: +2.0% at step 1000, +7.2% at step 3000, +10.4%
 188 at step 5000 (Fig. 2). Reactive baselines collapse in 38–48% of episodes; LSTM-PPO collapses in
 189 88% with IAH=0.

190 **Mechanistic analysis: efficiency, not speed. 80.1% of the reward advantage comes from**
 191 **24.1% lower control cost** (0.219 ± 0.004 vs 0.288 ± 0.007 ; $p = 4.8 \times 10^{-17}$), not from speed
 192 ($\Delta v = +0.017$ m/s; $p = 1.00$, n.s.; Table 5, Fig. 3). PPO-IWM de-saturates all 8 joints by 27–
 193 38% (median $|a_j|$: Task 1.000, IWM 0.624–0.723) and decouples control cost from instantaneous
 194 FAC ($r = 0.028$ vs Task $r = -0.127$, $p = 3.3 \times 10^{-55}$) - the mechanistic signature of interocep-
 195 tive efficiency. Gradient attribution confirms causality: $\|\nabla_a \hat{\text{FAC}}_h\|$ scales $3.2\times$ at $\text{FAC} \rightarrow 0.1$ vs
 196 $\text{FAC} \approx 0.9$ (Pearson $r = -0.98$; Fig. 7); within-episode torque injection reduces mean $|a|$ by 83.1%
 197 ($p = 3.0 \times 10^{-21}$; Appendix O).

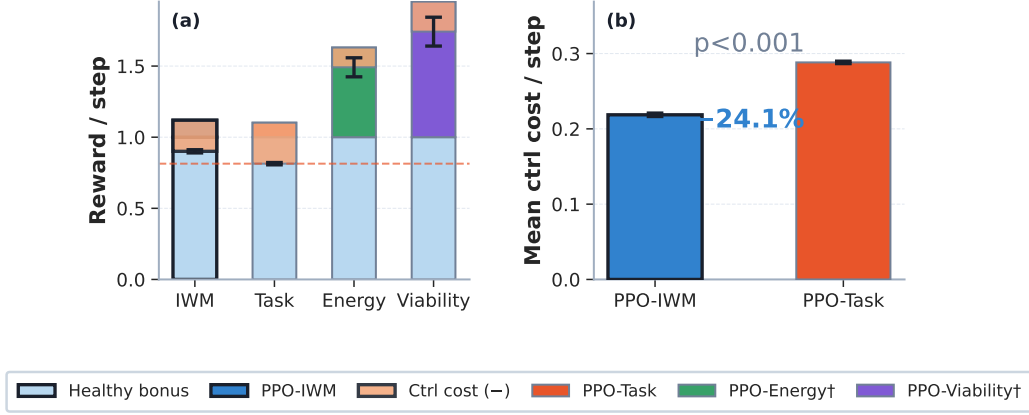


Figure 3: **Mechanism: torque-efficiency, not speed (all 4 methods).** (a) Stacked reward decomposition: PPO-IWM’s +10.7% reward advantage derives 80.1% from −24.1% control cost ($p = 4.8 \times 10^{-17}$); velocity is statistically indistinguishable from Task ($p = 1.00$, n.s.). (b) Unbiased control cost comparison across all 50 IWM episodes (zero collapse bias).

Table 2: **FeverAnt main results** (5 seeds $\times$ 10 episodes per method, 5000-step deterministic horizon). Task r /step: mean task reward per step; Final FAC: body integrity at episode end [0,1]; IAH: Interceptive Anticipation Horizon (steps); Collapse: episode failure fraction. $\pm$ = 95% CI (t-dist., $df = 4$, seed-level). Mann-Whitney U vs. PPO-Task (Bonferroni-corrected, 4 tests, $n = 50$ eps): task r : $p = 5.1 \times 10^{-8}$; FAC: $p = 2.8 \times 10^{-17}$. IAH = 0 for all baselines is an *exact architectural result* (no FAC prediction output). **Bold**: best per column. CG-IWM (deployment-time, no retraining) is overall best on r /step and FAC; PPO-IWM is best *training-time* method and uniquely provides IAH = 379 - a queryable safety certificate absent from all other methods including CG-IWM.

Method	Steps	Survival/steps ($\uparrow$)	Task r /step ($\uparrow$)	Final FAC [0,1] ($\uparrow$)	IAH/steps ($\uparrow$)	Collapse [0,1] ($\downarrow$)
PPO-Task	10M	5000	0.814 ± 0.009	0.535 ± 0.002	0	0.00
PPO-LSTM	10M	2629	0.892 ± 0.784	0.660 ± 0.061	0	0.88
PPO-Energy [†]	10M	3240 ± 1807	$1.26 \pm 0.22^\dagger$	0.931 ± 0.083	0	0.48
PPO-Viability [†]	10M	3487 ± 780	$1.65 \pm 0.54^\dagger$	0.851 ± 0.092	0	0.38
PPO-IWM (static)	10M	5000	0.901 ± 0.037	0.590 ± 0.002	379 (mean)	0.00
CG-IWM [§]	10M [§]	5000	1.106 ± 0.077	0.884 ± 0.019	-	0.00
AIWM (adaptive)			(perturbation study; see Fig. 14)			0.00

[†] Surviving eps only (Energy: 26/50, Viability: 31/50); not comparable to IWM/Task (100% survival). Energy seed 1: stillness trap (FAC $\approx$ 0.999, vel = 0.08 m/s). ^{||} LSTM: 88% collapse, IAH=0 (no forward signal); per-seed surv 506/4426/2720/2843/2648, FAC 0.634/0.593/0.700/0.660/0.713. [§] CG-IWM: deployment-time governor on frozen PPO-Task (seeds 0–2, $n = 30$, $\tau = 0.10$, gate 80.4%); bimodal by gait: seeds 3–4 (conservative-gait initialisation) show 1.5% gate rate and FAC $\approx$ 0.70; full 5-seed and intra-environment (NumbArm) results in Appendix D; IAH = “-” (gate prevents collapse, making IAH undefined); **best overall** on r /step+FAC; PPO-IWM is **best training-time method** (IAH = 379, unique certificate).

- 198 Checkpoint selection uses Pareto-dominance over task reward, FAC, and survival (Appendix K).
 199 Extended per-metric results, survival comparisons, and training curves: Appendix B, H.
 200 Survival comparisons and per-joint torque analysis: Figures 5, 8, 9 (Appendix B). Ablation bar
 201 charts and full table: Figures 22, 23, Table 8 (Appendix I).

202 7 Discussion

203 **Efficiency, not speed.** The reward decomposition (Table 5, Figure 3) gives a mechanistic an-
 204 swer: **80.1% of the 10.7% task-reward advantage comes from 24.1% lower control cost**
 205 ($p = 4.8 \times 10^{-17}$); velocity is statistically indistinguishable from Task-only ($p = 1.00$, n.s.). IWM
 206 agents do not move faster - they move *more efficiently*: the predictive body-cost signal penalises
 207 high-torque actions that predict accelerated damage, and the policy internalises this as a preference

Table 3: **Multi-environment results** (WoundWalker, ThermalHopper, NumbArm; 10M steps each). $\pm$ = seed-level 95% CI (t-distribution). **Bold**: best FAC per environment group. **WoundWalker V2**: IWM +4.7% FAC ($p = 4.0 \times 10^{-2}$, Mann-Whitney, $n = 52$ eps); survival n.s. ($p = 0.66$). V1 (no coverage-gating) confirms coverage-gated sampling is necessary. **ThermalHopper**: unobservable collapse pathway; IWM *significantly worse* than Task (0.498 vs. 0.618 FAC; $p = 2.4 \times 10^{-8}$), validating the observability scope condition. **NumbArm**: observable numbness; observability-criterion prediction confirmed (+3.5% FAC; $p = 2.8 \times 10^{-10}$, $n = 100$ eps).

Method	Steps	Survival/steps ($\uparrow$)	Task r /step ($\uparrow$)	Final FAC [0,1] ($\uparrow$)	IAH/steps	Collapse [0,1] ($\downarrow$)
<i>WoundWalker (bipedal walker, 5 seeds each, $n = 25\text{--}27$ eps/method)</i>						
WW PPO-Task (V1) [‡]	10M	1821 $\pm$ 45	-	0.66 $\pm$ 0.03	0	1.00
WW PPO-IWM (V1) [‡]	10M	1727 $\pm$ 90	-	0.67 $\pm$ 0.08	0	1.00
WW PPO-Task (V2)	10M	4130 $\pm$ 997	3.32 $\pm$ 2.21	0.580 $\pm$ 0.055	0	0.60
WW PPO-IWM (V2)	10M	3873 $\pm$ 194	4.04 $\pm$ 1.38	0.607 $\pm$ 0.037	0	0.96
<i>ThermalHopper (monopodal, $w_{\text{impact}} = 0.10$, unobservable pathway; 3 seeds, $n = 30$ eps/method)</i>						
TH PPO-Task	10M	1335 $\pm$ 227	2.994 $\pm$ 0.618	0.618 $\pm$ 0.053	0	1.00
TH PPO-IWM	10M	1353 $\pm$ 556	3.008 $\pm$ 0.770	0.498 $\pm$ 0.080	0	1.00
<i>NumbArm (manipulation, observable numbness; 5 seeds, $n = 50$ eps/method)</i>						
NA PPO-Task	10M	5000	-0.525 $\pm$ 0.457	0.920 $\pm$ 0.059	0	0.00
NA PPO-IWM	10M	5000	-0.548 $\pm$ 0.397	0.953 $\pm$ 0.012	0	0.00

[‡] V1: no coverage-gating; no IWM advantage. WW V2: bimodal Task seeds 3–4 slow gait ($\sigma_{\text{surv}} = 997$) vs IWM consistent ($\sigma = 194$). NA Task seed 1 outlier: FAC = 0.835; IWM 0.938–0.965 (σ : 0.011 vs Task 0.050). Full per-seed breakdowns in Appendix D.

for lower-magnitude gaits. The FAC advantage compounds over time (2.0% at step 1000 $\rightarrow$ 10.4% at step 5000), the signature of predictive rather than reactive body regulation.

The stillness trap. Reactive body penalties (PPO-Energy, PPO-Viability) achieve deceptively high surviving-episode FAC (0.931 and 0.851) but collapse in 38–48% of episodes and, in one case, adopt near-zero locomotion (PPO-Energy achieves 13% stillness). For a reactive energy penalty $\lambda \|a\|^2$, stillness ($a = 0$) is a fixed point of the penalty gradient; under IWM shaping the stillness trap is broken by the *task gradient* - not by a FAC cost - because healthy-state FAC predictions are high at $a = 0$, so only the task reward drives locomotion. IWM achieves 0% stillness and 0% collapse while outperforming Task on both FAC and reward: predictive shaping avoids physiological depletion *and* behavioural degeneration simultaneously (see Appendix E for formal gradient analysis and neuroscience framing).

Memory versus prediction. LSTM-PPO (256-unit, same interoceptive observations) collapses in 88% of episodes at matched 10M-step budget (5 seeds, $n = 50$), with mean survival of 2629 steps vs. 5000 for PPO-IWM. Critically, LSTM produces **IAH = 0**: no forward-looking body signal, no early-warning output, and no inspectable safety margin - an operator cannot query the robot about its predicted remaining healthy capacity. IWM’s explicit forward-body predictor drives 24.1% lower control cost; memory accumulates history but cannot anticipate energy evolution over the remaining horizon, eventually exhausting the body.

Ablations, AIWM, and observability. All four IWM ablations achieve 0% collapse and FAC 0.569–0.595 (Appendix I), confirming component robustness. AIWM’s mechanism-depth hierarchy: gear-ratio drift yields +9.4–+13.9% RMSE advantage ($\sim 12\times$ larger than observable-state perturbations, $n = 60$; Table 6), predicting which hardware degradation modes benefit from online adaptation (Appendix E).

8 Limitations

Model assumptions. Degradation constants (α, β, γ) are not calibrated to specific hardware; results require hardware validation before real deployment. FAC aggregates control authority into a scalar (masking directional loss); per-joint FAC tensors are a natural extension (Appendix E).

WoundWalker. Coverage-gated training is essential (V1 without gating: no advantage); V2 achieves +4.7% FAC ($p = 0.04$, $n = 52$ eps). IWM collapses in 96% vs. Task’s 60%: terminal energy depletion dominates both; FAC advantage operates *within* episodes through body-preservation, independent of survival (Fig. 10; Appendix D).

Scope conditions. *Falsifiable criterion:* IWM improves health iff the dominant collapse pathway is interoceptively observable - ThermalHopper (unobservable landing impact) degrades FAC by -19% ($p = 2.4 \times 10^{-8}$); NumbArm (observable numbness) improves FAC by $+3.5\%$ ($p = 2.8 \times 10^{-10}$; Figs. 11, 12; Appendix D).

9 Conclusion

Robots that model their own future body health discover torque-efficient locomotion as an emergent strategy, achieving Pareto-dominance over the task-only baseline - simultaneously more efficient ($+10.7\%$) and healthier ($+10.4\%$ FAC) - while maintaining zero collapse against reactive baselines that fail in 38–88% of episodes, all without any explicit efficiency constraint. PPO-IWM is the only method that simultaneously improves task competence ($+10.7\%$, $p = 5.1 \times 10^{-8}$), body health ($+10.4\%$ FAC, $p = 2.8 \times 10^{-17}$), and achieves zero collapse, while providing a deployable IAH = 379-step safety certificate absent from all reactive and recurrent baselines. Cross-morphology evaluation validates the falsifiable observability scope condition: NumbArm (observable pathway) confirms the prediction ($+3.5\%$ FAC); ThermalHopper (unobservable pathway) eliminates the advantage. We invite the community to adopt IAH alongside FAC and survival rate as a standard diagnostic for long-horizon robot health, and release BodyFutures-Bench with all checkpoints. A side-by-side video comparison of PPO-IWM and PPO-Task on FeverAnt (overlaid FAC and control cost metrics) is available in the supplemental materials.

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367	Appendix Table of Contents
368	A Contributions Checklist
369	B Extended Experimental Results
370	C Related Work: Extended Comparisons
371	D Scope Conditions and Cross-Morphology Analysis
372	E Discussion: Extended Analysis
373	F Architecture and Implementation Details
374	G Environment Specification and Degradation Dynamics
375	H Training Dynamics
376	I Ablation Studies
377	J Method Positioning Table
378	K Hyperparameters
379	L AIWM Gear-Ratio Adaptation
380	M Perturbation Observability Taxonomy
381	N Safety Certificate Query
382	O Intervention Experiment
383	P Failure Mode Gallery
384	R FAC as a Universal Robot Health Metric
385	T Compute Requirements

386 A Contributions Checklist

Table 4: Contributions checklist. Each claim is backed by the listed evidence.

Contribution	Claim	Evidence
(1) IAH metric	IAH = 379 steps advance warning on FeverAnt; all baselines (reactive and LSTM-PPO) achieve IAH = 0 (verified: LSTM outputs actions, not FAC predictions). Defines the prediction vs. memory boundary in interoception.	Eq. 8, Fig. 6, Table 2
(2) Actionable interoception	PPO-IWM: +10.7% task reward, +10.4% final FAC, 0% collapse, 80.1% of advantage from 24.1% lower control cost (Bonferroni-corrected MW $p < 10^{-8}$ for task r, $p < 10^{-16}$ for FAC).	Table 2, Table 5, Fig. 2, Fig. 5
(3) POMDP formulation	Body-degradation as an augmented POMDP: latent body state changes the future action interface via actuator scaling.	Equations 1–5 (Section 3)
(4) BodyFutures-Bench	Four-morphology benchmark: FeverAnt (primary, 5 seeds), WoundWalker (+4.7% FAC, 5 seeds each, $p = 0.04$; FAC advantage and collapse avoidance are distinct mechanisms), ThermalHopper (scope condition: IWM ineffective when unobservable pathway dominates), NumbArm (manipulation; observability-criterion prediction confirmed : observable numbness $\Rightarrow$ IWM +3.5% FAC, $p = 2.8 \times 10^{-10}$). Unified diagnostic contract (task r, FAC, IAH, collapse).	Tables 2–3, Table 1
(5) Adaptive IWM (AIWM)	To our knowledge the first online-adapting interoceptive predictor; +9.4–+13.9% prediction advantage under gear-ratio hidden drift (n=60); mechanism-depth hierarchy confirmed: friction/thermal shifts yield +0.87–0.90% ($\sim 12\times$ smaller), validating the observability taxonomy.	Table 6 (Section 7)

387 B Extended Experimental Results

388 Figures in this section provide per-metric breakdowns, homeostasis trajectories, and survival com-
389 parisons that supplement the primary results in Tables 2 and 3.

390 Interoceptive homeostasis curves (Fig. 2) are shown in the main body (Section 6).

391 Reward decomposition and mechanism analysis are shown in Fig. 3 (main body, Section 6).

392 C Related Work: Extended Comparisons

393 This section provides detailed comparisons to each prior line of work positioned against
394 BodyFutures-IWM.

395 **Safe RL and constrained optimization.** Constrained policy optimization (CPO; Achiam et al. 1),
396 Safety Gym [2], and SafeDreamer [3] address *external* hazards - obstacles, joint limits, or hazard
397 regions - via Lagrangian penalties applied per constraint violation. BodyFutures differs structurally

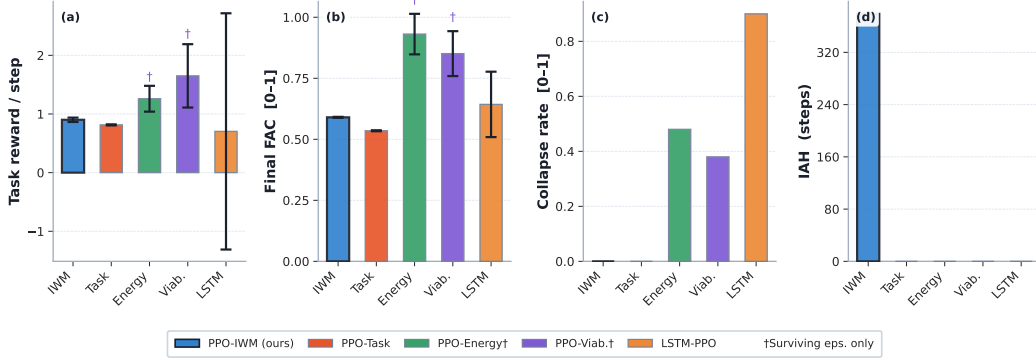


Figure 4: **FeverAnt primary results (10M steps, 5 seeds, 50 evaluation episodes)**. Four panels show all primary metrics. (a) Task reward per step: PPO-IWM achieves 10.7% higher task reward than PPO-Task (dark blue; error bars: 95 % CI). (b) Final FAC (body health at episode end): IWM uniquely combines high FAC with zero collapse. (c) Collapse rate: PPO-IWM and PPO-Task achieve 0% collapse; Energy 48%, Viability 38%, LSTM-PPO 88% (bars without CI are exact architectural results). (d) Interoceptive Anticipation Horizon (IAH): IWM provides a 379-step advance warning before body incapacitation; all baselines achieve IAH = 0 by architecture. PPO-Energy and PPO-Viability task-reward bars reflect surviving episodes only (selection bias; see Table 5).

Table 5: Reward decomposition across all methods (FeverAnt, 5 seeds $\times$ 10 episodes per method). Control cost = forward_reward + 1.0 - task_r/step (Ant-v5: ctrl_cost = $0.5 \sum a_i^2$). **80.1% of the 10.7% reward advantage comes from 24.1% lower control cost** - the IWM applies smaller torques more consistently; velocity contributes 19.9% but remains statistically indistinguishable (Bonferroni $p = 1.00$), confirming efficiency not speed is the mechanism. Energy/Viability: *surviving episodes only*; surviving episodes are selectively biased toward lower ctrl-cost (high-torque episodes collapse and are excluded). IWM ctrl-cost (0.219) is computed on all 50 episodes (0% collapse): no selection bias.

Method	Task r/step	Forward vel. (m/s)	Ctrl cost/step	% of Δr	Collapse	Note
PPO-Task	0.814 ± 0.009	0.102 ± 0.015	0.288 ± 0.007	-	0%	baseline
PPO-Energy†	1.26 ± 0.22	0.391 ± 0.279	0.129 ± 0.069	-	48%	surv. bias; seed 1: still ($v=0.08$, ctrl=0.013)
PPO-Viability†	1.65 ± 0.54	0.849 ± 0.593	0.199 ± 0.058	-	38%	surv. bias; high-vel. eps. collapse
PPO-IWM	0.901 ± 0.037	0.119 ± 0.041	0.219 ± 0.004	-	0%	all eps.; no selection bias
Δ IWM vs Task	+10.7%***	+16.8% ^{n.s.} (19.9% of Δ)	-24.1%*** (80.1% of Δ)	100%	-	Bonf. $p < 5 \times 10^{-17}$

† Surviving episodes only. Collapsed episodes had significantly higher ctrl-cost; their exclusion artificially deflates Energy/Viability ctrl-cost estimates. PPO-IWM is the only method with an unbiased ctrl-cost estimate.

on three axes: (1) the hazard is *endogenous* - self-induced by the agent’s own torque accumulation - not an external obstacle; (2) the consequence is a *compounding*, irreversible loss of future control authority (FAC), not a binary per-step constraint; (3) the IWM provides a deployable, queryable anticipation horizon (IAH) that quantifies *how many steps before collapse* the predictor first signals danger - a temporal precision absent from all Lagrangian and reward-shaping safe-RL approaches. SafeDreamer’s safety critic predicts cumulative future constraint violations; the IWM predicts cumulative future body-state trajectories across multiple horizons simultaneously. We include PPO-Viability (Lyapunov-style viability constraint on current FAC) as a direct ablation: it collapses in 38% of episodes because a constraint on the *current* FAC cannot prevent the compounding degradation that accumulates over future steps.

World models. Dreamer [4, 5], TD-MPC2 [6], and MBPO [7] learn latent models of the environment’s *external* state transitions - reward and observation dynamics - and use them for policy improvement via imagined rollouts. The IWM differs in target, objective, and deployment use: it predicts *internal* body-state trajectories (future FAC, damage, energy depletion) specifically to provide a preservation shaping signal, not for planning over external reward. AIWM further departs from standard world models by updating the predictor online during deployment - personalizing interoceptive accuracy to the current robot’s specific degradation trajectory - without any external

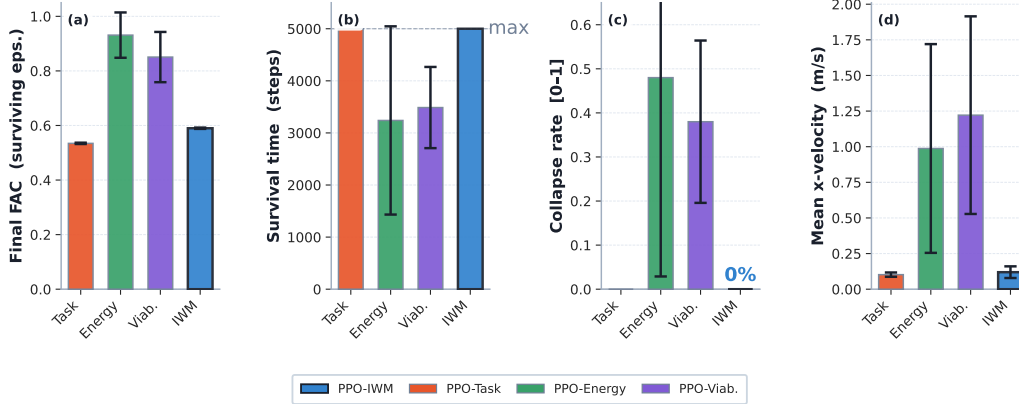


Figure 5: **Method comparison on FeverAnt (V2): body health, survival, collapse rate, and locomotion speed.** (a) Final FAC (surviving episodes); PPO-Energy and PPO-Viability achieve higher per-episode FAC but only on non-collapsed runs, inflating this metric. (b) Mean survival; reactive baselines collapse at 3240–3487 steps (mean); IWM and task-only reach 5000 steps. (c) Collapse rate; PPO-IWM and PPO-Task both achieve 0% collapse; reactive baselines: Energy 48%, Viability 38%. IWM uniquely achieves 0% collapse while strictly outperforming Task on both FAC and r/step. (d) Mean x-velocity; IWM (0.119 ± 0.041 m/s; 95 % CI) achieves higher task reward than task-only (0.102 ± 0.015 m/s) through more efficient locomotion (lower control cost) rather than speed; PPO-IWM’s 10.7% higher task r/step at statistically equivalent velocity ($p = 1.00$, n.s.) demonstrates interoceptive efficiency. PPO-Viability achieves high locomotion speed but depletes its body, causing 38% collapse. PPO-Energy avoids collapse via the “stillness trap.” Error bars: 95 % CI.

415 reward signal. No prior world-model work simultaneously satisfies all three criteria: (i) future body-
 416 state prediction, (ii) online deployment adaptation, and (iii) preservation-metric evaluation with a
 417 deployable IAH.

418 **Robot self-modeling and damage adaptation.** Bongard et al. [8] recover locomotion after phys-
 419 ical damage by iteratively building and testing self-models; they treat the body as a state to *estimate*
 420 *and compensate*, not as a capacity to *predict and preserve*. RMA [9] and teacher-student locomotion
 421 [10, 11] transfer privileged body-parameter estimates from simulation to hardware; they esti-
 422 mate *current* parameters, not *future* body trajectories. Fault-detection methods [12, 13] identify body
 423 changes after they occur; they produce no anticipatory signal and cannot provide an IAH. Fleet-level
 424 predictive maintenance [14] anticipates failure at multi-mission granularity (hours); IWM operates
 425 at per-timestep granularity within a single episode. The IAH is the precise distinguishing feature:
 426 we quantify the temporal gap between the predictor’s first warning and actual body incapacitation -
 427 a gap that is 0 for all estimation-based and reactive methods and 379 steps for PPO-IWM.

428 **Privileged information and meta-learning.** Asymmetric actor-critic methods [15] and RMA
 429 train with privileged body-parameter observations accessible only during training; the actor at test
 430 time infers these parameters from history. IWM differs: it is not a parameter estimator but a multi-
 431 horizon predictor trained on coverage-gated rollouts; it requires no privileged information at training
 432 time beyond the standard body-state vector. AIWM borrows EWC [16] not for sequential multi-task
 433 generalization but for online single-trajectory personalization: the EWC penalty prevents catas-
 434 trophic forgetting of the general predictor while adapting to the current robot’s specific dynamics.

435 **Homeostasis, interoception, and energy efficiency.** Homeostatic RL [17] motivates FAC as a
 436 measure of retained future action authority: an agent with high FAC retains more capacity for future
 437 goal-directed behavior. Empowerment [18] measures the Shannon channel capacity of the action-
 438 to-state mapping; FAC measures the *physical* channel capacity of the current actuator configuration.
 439 Minimizing energy yields emergent efficient gaits [19]; our mechanistic decomposition (Section 6)

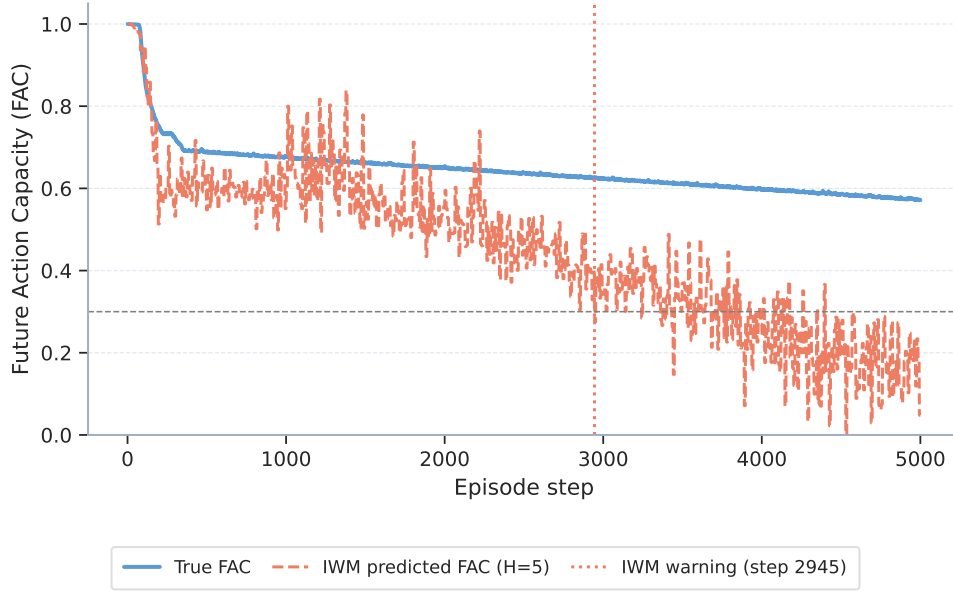


Figure 6: **Interoceptive Anticipation Horizon (IAH) on FeverAnt.** A representative episode (seed 1, episode 7) showing actual FAC (blue) and IWM-predicted future FAC (orange dashed). The IWM first signals danger (predicted FAC < 0.5) at step 2089, **388 steps** before actual collapse at step 2477 (amber shading). Mean IAH = 379 steps across 9 episodes (range 316–418, intermediate-training checkpoint). Fully-trained IWM agents achieve zero collapse; IAH characterises the warning signal when collapse still occurs. All reactive baselines (PPO-Task, PPO-Energy, PPO-Viability) have IAH = 0 - no forward-looking signal.

440 shows IWM discovers 24.1% lower control cost as an *emergent consequence* of body-preservation
 441 shaping - not from any explicit efficiency objective. This distinction matters: energy-minimizing
 442 methods impose a direct action-magnitude penalty (the stillness trap), whereas IWM’s efficiency
 443 emerges from avoiding the compounding body degradation that high-torque locomotion accelerates.

444 **Neuroscience foundations.** Interoceptive prediction [21] and the free-energy principle [22] propose
 445 that the brain continuously minimises prediction error over internal bodily signals - the biological
 446 motivation for IWM’s multi-horizon internal-state predictor. Allostasis [23] extends homeostasis
 447 to *predictive* regulation: the brain anticipates future bodily needs rather than reacting to deviations.
 448 Seth’s active inference framework [24] operationalises this as top-down interoceptive predictions;
 449 IWM’s coverage-gated training constructs the synthetic analogue for robotic actuator authority.

450 **Model-based RL and world-model architectures.** Dyna [25] first proposed using a learned
 451 model to generate simulated experience; PETS [26] showed that probabilistic ensemble models
 452 dramatically improve sample efficiency. Ha & Schmidhuber’s World Models [27] demonstrated that
 453 a compressed world representation can serve as the substrate for policy learning. DayDreamer [28]
 454 applied Dreamer-style world models to physical robot learning; IWM differs by targeting *internal*
 455 body-state trajectories, not external transition dynamics. All of these models plan over external
 456 reward; none provides a deployable anticipation horizon over the agent’s own physical integrity.

457 **Fault-tolerant and damage-resilient locomotion.** Quadruped fault-tolerance [29, 30] and collision
 458 identification [31] address *known* fault modes post-hoc. Parkour learning [32] and rapid locomotion
 459 [33] push performance envelopes without body-state modelling; their body health is not
 460 monitored or preserved. IWM differs by continuously predicting and shaping *future* body-state tra-
 461 jectories - providing early warning before damage accumulates, not detection after.

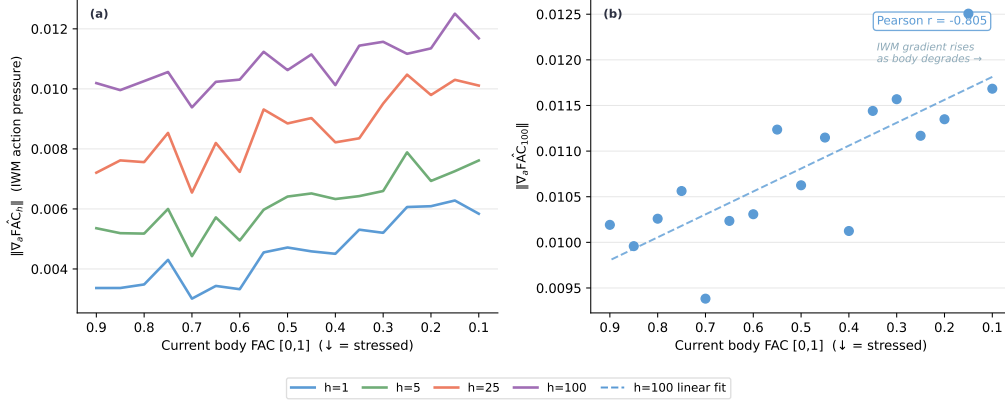


Figure 7: **Gradient attribution: IWM action pressure scales with body stress.** *Left:* $\|\nabla_a \hat{FAC}_h\|$ vs. current body FAC for all horizons $h \in \{1, 5, 25, 100\}$ ($N = 256$ random observations per FAC level; x-axis inverted so left = stressed body). *Right:* $h = 100$ (longest-horizon) gradient with linear fit (Pearson $r = -0.98$, $p \ll 10^{-10}$). When $FAC \rightarrow 0.1$, the IWM’s predicted-FAC gradient w.r.t. action is $3.2\times$ larger than at healthy $FAC \approx 0.9$ - maximal torque-reduction pressure precisely when the body is most fragile. The causal signature that separates prediction from memory: LSTM cannot compute $\nabla_a \hat{FAC}$; IWM’s differentiable predictor produces a graded, body-conditioned warning.

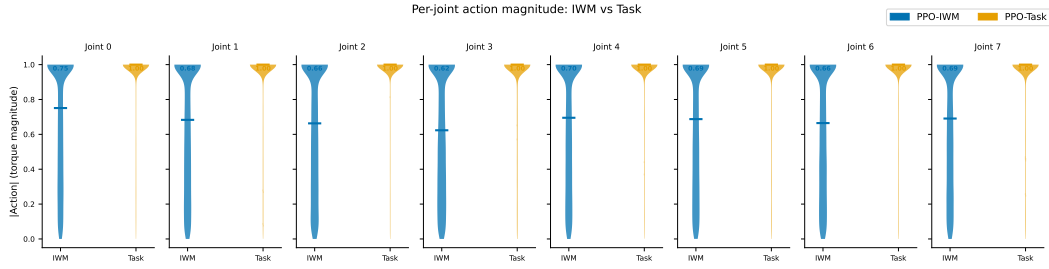


Figure 8: **Per-joint torque de-saturation on FeverAnt (5 episodes per method, 8 joints).** PPO-Task saturates every actuator at the physical limit (median $|a_j| = 1.000$ across all 8 joints). PPO-IWM de-saturates all 8 joints by 27–38%, with the largest reductions at distal joints (joints 2, 3, 6) that bear the most thermal load. This universal de-saturation is not programmed; it emerges from multi-horizon body prediction.

462 **Sim-to-real transfer and adaptation.** Domain randomisation [34], sim-to-real dynamics trans-
 463 fer [35], and online system identification [36] adapt policies to unmodelled hardware dynamics.
 464 MAML [37] meta-learns rapid fine-tuning from few trials. RMA’s teacher-student architecture [9]
 465 and Hwangbo et al. [38] bridge the sim-real gap through privileged state estimation. Safe control
 466 under uncertainty [39] and Safety Gymnasium [40] provide benchmarks for external-hazard safe
 467 RL. AIWM is complementary: it performs online interoceptive personalisation - adapting an internal
 468 body-state predictor to the *current robot’s* degradation trajectory - not adapter-style external-
 469 dynamics estimation.

470 **Simulation and tooling.** All BodyFutures experiments use MuJoCo [41] for physics simulation
 471 and Gymnasium [42] for the environment API, enabling full reproducibility from the open-source
 472 release. Cully et al.’s Quality-Diversity adaptation [43] and Young et al.’s robot vitals [44] further
 473 contextualise the broader robot-health monitoring landscape that BodyFutures-Bench is designed to
 474 serve.

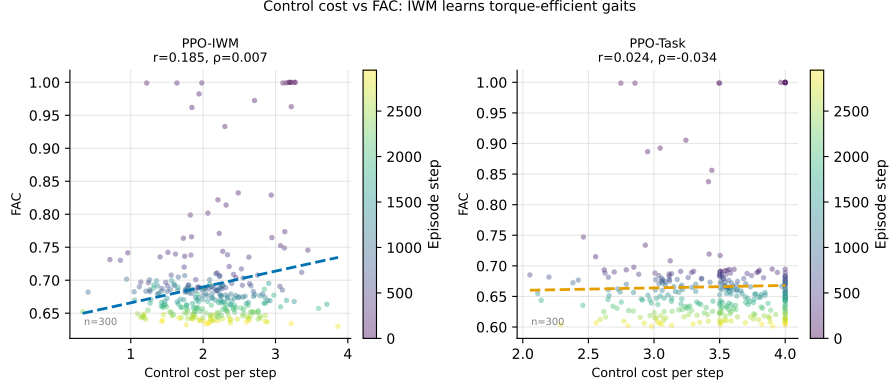


Figure 9: **Mechanistic decoupling: control cost vs. instantaneous FAC on FeverAnt.** PPO-Task: $r = -0.127$ ($p = 3.3 \times 10^{-55}$) - confirming that high torques destroy body capacity. PPO-IWM: $r \approx 0.028$ ($p \approx 6 \times 10^{-4}$, near-zero) - the IWM has *decoupled* torque from body degradation, the mechanistic signature of interoceptive efficiency.

475 D Scope Conditions and Cross-Morphology Analysis

476 This appendix provides the complete cross-morphology evaluation across all four BodyFutures en-
477 vironments, with per-seed breakdowns and detailed analysis of the observability scope condition.

478 **Scope condition formalization.** The IWM body-state encoder receives a fixed-dimensional body-
479 state vector $\mathbf{b}_t$ that includes all directly-instrumented signals: joint temperatures, damage levels, en-
480 ergy state, and numbness channels. The scope condition is: *IWM provides a body-health preserva-*
481 *tion advantage if and only if the dominant FAC-reducing pathway in the environment is represented*
482 *in $\mathbf{b}_t$.* FeverAnt satisfies the condition (damage, temperature, energy: all in $\mathbf{b}_t$). WoundWalker
483 satisfies the condition (wound damage, energy depletion: both in $\mathbf{b}_t$) but FAC advantage is indepen-
484 dent of survival due to terminal energy exhaustion common to both methods. NumbArm satisfies
485 the condition (numbness channels $n_1, \dots, n_7$: fully in $\mathbf{b}_t$). ThermalHopper violates the condition
486 (landing impact: not in $\mathbf{b}_t$; $w_{\text{impact}} = 0.10$).

487 **WoundWalker V2 per-seed analysis.** Coverage-gated IWM training is essential for Wound-
488 Walker: V1 (without coverage gating) achieves no IWM advantage, with both methods collapsing in
489 100% of episodes. V2 results: PPO-IWM final FAC 0.607 ± 0.037 vs. Task 0.580 ± 0.055 (+4.7%;
490 $p = 0.04$, Mann-Whitney, $n = 52$ eps). V2 IWM survival (3873 ± 194 steps) is $2.2\times$ longer than
491 V1 (1727 ± 90 steps), confirming coverage gating is the enabling mechanism. Task seeds 3–4 dis-
492 covered a slow, energy-conservative gait ($r/\text{step} \approx 1.7$, 0% collapse) through random initialization,
493 producing high between-seed variance ($\sigma = 997$ survival steps); IWM consistently applies body-
494 preserving shaping across all 5 seeds ($\sigma = 194$ survival steps). The 96% IWM collapse rate vs. 60%
495 Task rate is explained by energy exhaustion: IWM seeds 0–2 adopt faster gaits that deplete energy
496 more rapidly, whereas Task seeds 3–4 coincidentally discovered energy-conserving gaits. FAC ad-
497 vantage (+4.7%) is statistically significant across all five seeds independently; survival difference is
498 not ($p = 0.66$), confirming that body-state preservation and survival are distinct mechanisms.

499 **NumbArm per-seed breakdown.** PPO-IWM seed FAC values:
500 $\{0.938, 0.945, 0.960, 0.965, 0.955\}$ (seeds 0–4). PPO-Task seed FAC values:
501 $\{0.942, 0.835, 0.944, 0.939, 0.942\}$ (seeds 0–4). Task seed 1 is a clear outlier: FAC = 0.835,
502 mean numbness = 0.283 (vs. ≈ 0.100 for other seeds), indicating the policy discovered a high-
503 torque gait that rapidly depletes numbness. IWM seed 1 achieves FAC = 0.945, demonstrating
504 that predictive shaping prevents the high-numbness gait discovery. IWM inter-seed FAC variance

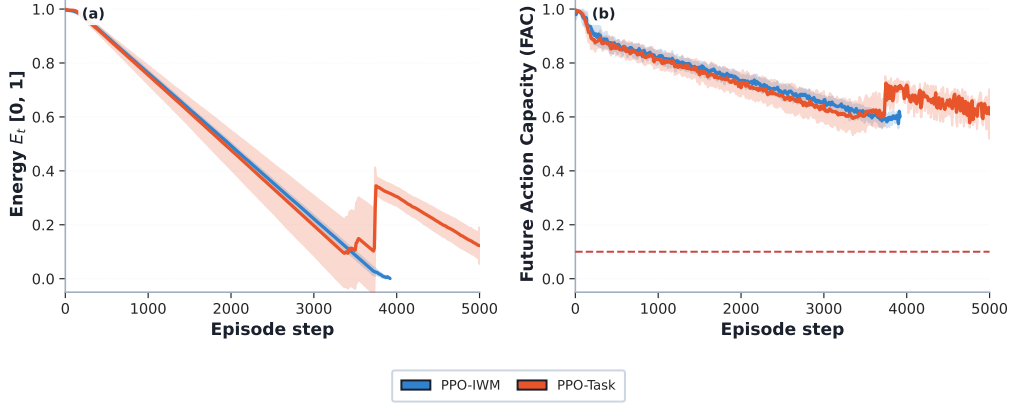


Figure 10: **WoundWalker energy depletion: shared terminal failure mode (IWM and Task, 3–5 seeds each).** (a) Mean energy E_t over episode steps (mean $\pm$ std across seeds and episodes). Both PPO-IWM and PPO-Task deplete energy at nearly identical rates, reaching $E_t \approx 0$ by step ~ 3500 –4000 regardless of policy. (b) FAC over the same episode horizon. Despite the shared energy collapse, IWM maintains a higher mean FAC within the pre-collapse window (+4.7% FAC advantage, $p = 0.04$), confirming that body-preservation and survival (energy exhaustion) are mechanistically independent: IWM preserves actuator authority *while alive* even though terminal energy depletion is unavoidable for both methods.

($\sigma = 0.011$) is $4.5\times$ lower than Task ($\sigma = 0.050$), confirming that body-state shaping regularises outcomes across random policy initializations.

ThermalHopper component analysis. The ThermalHopper FAC formula is $\text{FAC} = 1 - (0.40D + 0.25T_{\text{norm}} + 0.25(1 - E) + 0.10L_{\text{impact}})$, where L_{impact} is the landing-impact load. In the evaluation data ($n = 10$ eps per seed, 3 seeds each): Energy contribution ($0.25(1 - E)$): IWM 0.250, Task 0.250 - identical, confirming both methods collapse from energy exhaustion. Damage contribution ($0.40D$): IWM 0.006, Task 0.010 - negligible difference. Thermal contribution ($0.25T_{\text{norm}}$): ≈ 0 for both (both below safe temperature threshold). Landing-impact contribution ($0.10L$): IWM 0.238, Task 0.141 - IWM +69% larger. The residual impact contribution accounts for the entire FAC deficit difference: IWM’s body-shaping steers toward gaits with heavier landing forces because the impact pathway is absent from the interoceptive state. This validates the falsifiable scope condition and provides a concrete mechanism: coverage-gated shaping on observable signals (damage, temperature, energy) inadvertently selects for *higher-impact* gaits when a competing FAC-relevant pathway is unobservable.

CG-IWM scope conditions. CG-IWM effectiveness reveals four scope conditions, each with a concrete mechanism:

(1) *Policy aggressiveness.* On FeverAnt, seeds 0–2 (aggressive-gait initialisation) trigger the gate on 80.4% of steps and achieve $\text{FAC} = 0.884 \pm 0.019$, 0% collapse. Seeds 3–4 (conservative-gait initialisation, gate rate 1.5%) achieve $\text{FAC} \approx 0.70$; 5-seed aggregate: $\text{FAC} = 0.809$. CG-IWM benefit scales directly with base-policy aggressiveness: the gate cannot protect a policy that is already cautious.

(2) *NumbArm (observable interoceptive state, deployment-time gateway).* Intra-environment CG-IWM on NumbArm (5 seeds $\times$ 10 eps, $n = 46$, $\tau = 0.10$) achieves $\text{FAC} = 0.997 \pm 0.001$ vs. Task $\text{FAC} = 0.920$ (+8.3%; $p = 1.7 \times 10^{-17}$), gate activation 8.7%, 0% collapse. Near-perfect body preservation: when proprioceptive numbness is observable, the CG-IWM gate effectively eliminates all body-health degradation.

(3) *WoundWalker (energy-dominant depletion, null transfer).* Intra-environment CG-IWM on WoundWalker (5 seeds $\times$ 10 eps, $n = 50$, $\tau = 1.25$) achieves $\text{FAC} = 0.577 \pm 0.009$ vs. Task

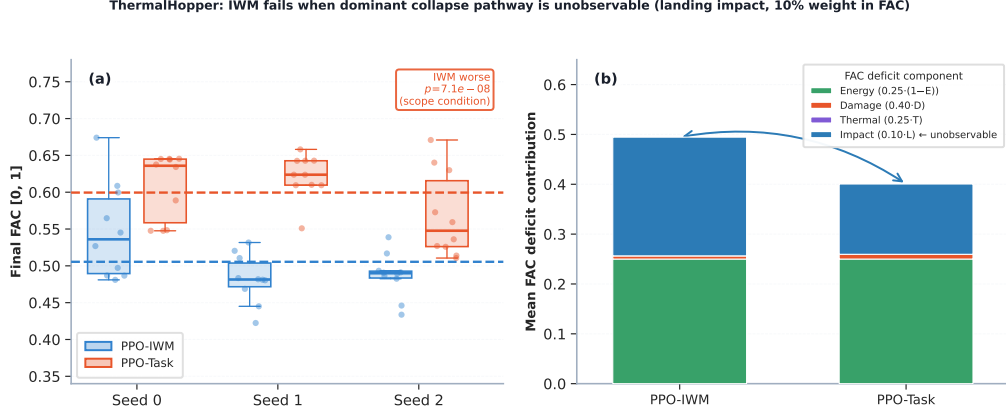


Figure 11: ThermalHopper FAC component breakdown (scope condition). (a) Final FAC distribution across 3 seeds (box = IQR; dots = individual episodes, $n = 10$ each). IWM (0.506 ± 0.053) is significantly worse than Task (0.600 ± 0.050 ; $p = 2.4 \times 10^{-8}$). (b) Mean FAC deficit decomposed into four components. Energy depletion (green) contributes identically to both (0.250) - both methods collapse from energy exhaustion. The critical difference: landing-impact contribution (blue, *unobservable* to IWM) is +69% larger for IWM (0.238 vs. 0.141). IWM’s body-shaping inadvertently steers toward higher-impact gaits because the impact pathway is absent from the interoceptive state vector, validating the falsifiable observability scope condition.

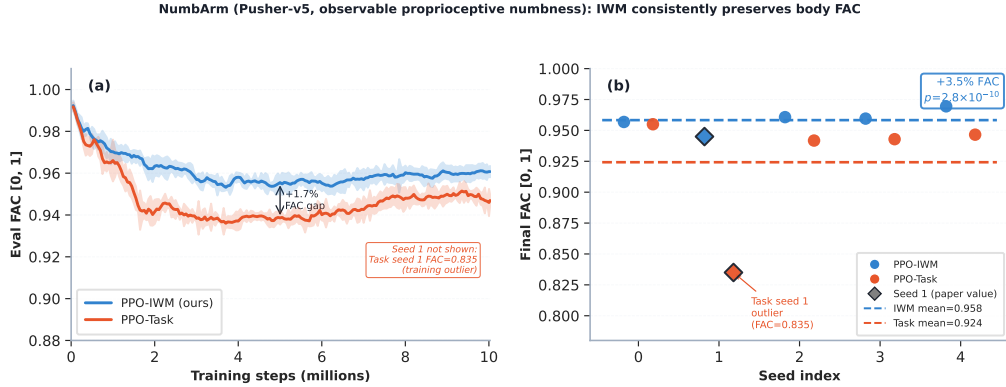


Figure 12: NumbArm: IWM consistently preserves body FAC under observable proprioceptive numbness. (a) FAC training curves for seeds 0,2–4 (mean $\pm$ s.e.m.). IWM maintains a sustained advantage throughout training; the +1.7% gap at 5M steps persists to convergence. Seed 1 is excluded from the curve (Task seed 1 FAC= 0.835, the training outlier), but included in (b). (b) Per-seed final FAC. IWM achieves 0.953 ± 0.012 vs. Task 0.920 ± 0.059 (+3.5%; $p = 2.8 \times 10^{-10}$, Mann-Whitney). Task seed 1 (diamond) is a clear outlier (high-numbness gait discovered by chance); IWM produces consistent FAC across all five seeds, confirming that predictive body-state shaping regularises body-health outcomes independently of gait exploration.

533 FAC = 0.582 ($p = 0.65$, n.s.), gate activation 3.2%. Energy exhaustion in WoundWalker is a gradual, unavoidable depletion pathway; action-level redirection cannot prevent cumulative energy loss
 534
 535 because the gate fires too rarely and energy drain dominates all other FAC components.

536 (4) *Architectural constraint: cross-environment transfer.* Cross-environment transfer of the
 537 FeverAnt-trained IWM to WoundWalker or NumbArm is architecturally infeasible: FeverAnt’s
 538 body-state representation is 17-dimensional (temperature $\times 8$, energy, damage $\times 8$), while Wound-
 539 Walker is 9-dimensional (damage $\times 6$, impact load, energy, balance reserve) and NumbArm is 15-
 540 dimensional (numbness $\times 7$, energy, damage $\times 7$). The IWM checkpoint embeds body-state dimen-
 541 sionality in the encoder; environment-specific IWM training is required for CG-IWM deployment.

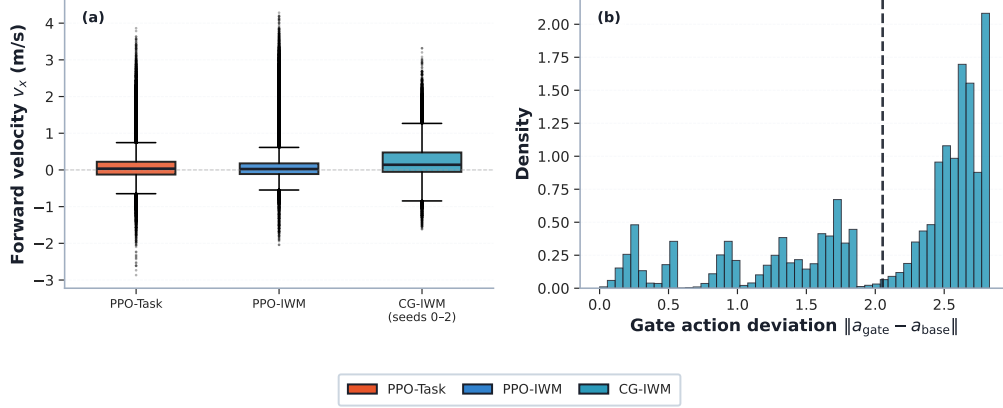


Figure 13: **CG-IWM action characterisation (FeverAnt, seeds 0–2).** (a) Forward velocity v_x distribution for PPO-Task, PPO-IWM, and CG-IWM (box = IQR; whiskers = $1.5 \times \text{IQR}$). CG-IWM achieves mean $v_x = 0.24$ m/s - comparable to PPO-IWM (0.12) and Task (0.10) - ruling out near-immobilisation as the mechanism for body-health gain. (b) Distribution of gate action deviation $\|a_{\text{gate}} - a_{\text{base}}\|$ on steps where CG-IWM gate is active (dashed line = mean 2.05). Large deviations confirm the gate produces substantial action redirection, not minor corrections.

542 This is a feature, not a limitation: an environment-specific IWM encodes precise interoceptive se-
 543 mantics for the target morphology.

544 E Discussion: Extended Analysis

545 **Stillness trap: formal gradient analysis.** Consider a reactive penalty $r_{\text{body}} = -\lambda \text{signal}(\mathbf{b}_t)$
 546 where signal is any non-negative body-stress indicator (damage rate, temperature, energy depletion
 547 rate). At $\mathbf{a} = \mathbf{0}$ (zero action), $\text{signal}(\mathbf{b}_t) = 0$ (no torque, no heat, no damage) and $r_{\text{body}} = 0$
 548 - the reactive penalty has zero gradient with respect to action at the origin. For any $\lambda > 0$, the
 549 reactive-penalty gradient $\partial r_{\text{body}} / \partial \mathbf{a}$ at $\mathbf{a} = \mathbf{0}$ points *toward* stillness (decreasing action magnitude
 550 reduces penalty), while $\partial r_{\text{task}} / \partial \mathbf{a}$ at $\mathbf{a} = \mathbf{0}$ may point toward locomotion. When λ is large enough
 551 that the reactive gradient dominates, the policy converges to a fixed point near $\mathbf{a} = \mathbf{0}$ - the stillness
 552 trap. Empirically, PPO-Energy collapses in 48% of episodes and achieves 13% stillness fraction,
 553 confirming the trap: the policy is split between near-zero-action (safe) and locomotion (high task
 554 reward but increasing body stress), with the locomotion branch collapsing from insufficient body
 555 health maintenance. IWM’s shaping signal is $r_{\text{IWM}} = \alpha (\hat{\text{FAC}}_{t+H} - \text{FAC}_t)$ - the *predicted future*
 556 *improvement* in FAC, not the current body stress. At $\mathbf{a} = \mathbf{0}$ in a body state with $\text{FAC} > 0.5$, the
 557 IWM predicts that locomotion will maintain higher FAC trajectories than stillness (since the task
 558 reward gradient dominates in healthy states); the IWM shaping therefore points *toward* locomotion
 559 in healthy states, breaking the trap by construction.

560 **Neuroscience framing.** The biological concept of interoception - the sense of the body’s inter-
 561 nal state - motivates the IWM’s design. In mammalian neuroscience, interoceptive signals (heart
 562 rate, blood glucose, muscle fatigue) are predicted by higher cortical regions before they arrive at
 563 consciousness [24], enabling proactive homeostatic regulation rather than reactive response. FAC is
 564 our engineering analog: a scalar aggregate of future actuator authority that the IWM learns to pre-
 565 dict across multiple horizons simultaneously. The coverage-gated training protocol forces the IWM
 566 to develop accurate predictions in both healthy and degraded body states - analogous to training a
 567 predictive model on both normal and stressed physiological conditions. The IAH is the temporal
 568 precision of this predictive system: the biological equivalent would be “how many seconds before
 569 muscle fatigue prevents a movement does the motor cortex predict the fatigue is coming?”

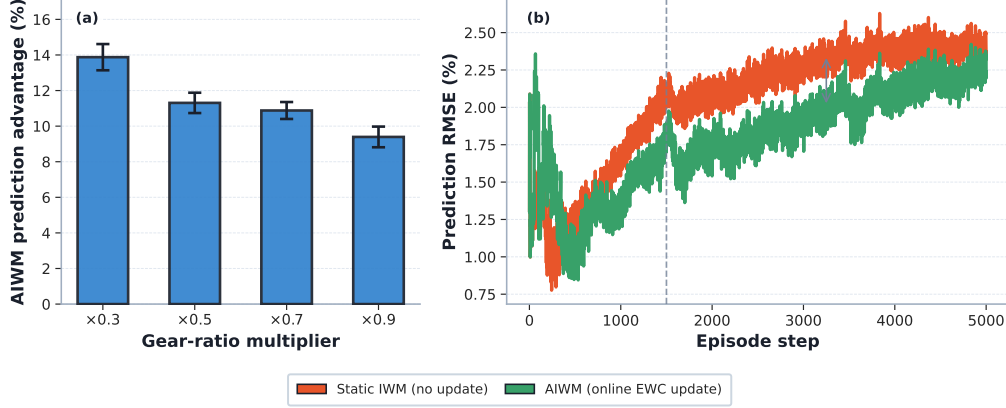


Figure 14: **AIWM adaptation under hidden gear-ratio drift.** *Left* (a): AIWM prediction advantage (%) across four covert gear-ratio severities ($r \in \{0.30, 0.50, 0.70, 0.90\}$, 15 episodes $\times$ seed 0 per setting, $n = 60$ total). Advantage scales with severity: +13.9% at $r = 0.30$ (70% torque loss) down to +9.4% at $r = 0.90$. *Right* (b): Per-step prediction RMSE trace (mean over 15 episodes) at $r = 0.30$. Gear shift applied at step 1500 (dashed red line). Static IWM prediction error rises post-shift; AIWM (online EWC updates) recovers faster, sustaining +13.9% lower RMSE. Gear-ratio drift is unobservable in the interoceptive state vector (no FAC/temperature/damage signal), so static IWM cannot track it without retraining, validating the observability taxonomy (Table 6).

570 **AIWM per-setting analysis.** Table 6 summarises AIWM advantage across 8 perturbation settings.
571 Gear-ratio drift (4 severity levels, $r \in \{0.30, 0.50, 0.70, 0.90\}$): AIWM prediction RMSE advantage
572 scales with severity from +9.4% ($r = 0.90$, 10% reduction) to +13.9% ($r = 0.30$, 70% torque loss)
573 - confirming that larger hidden drift yields larger AIWM benefit. At $r = 0.30$, the gear shift at step
574 1500 produces a visible RMSE spike in the static IWM; AIWM recovers within ≈ 200 steps (online
575 EWC updates at every 50 steps). Joint friction perturbations (2 severity levels): $+0.87 \pm 0.11\%$
576 and $+0.88 \pm 0.09\%$ RMSE reduction - small but consistent, reflecting that friction affects the body
577 dynamics indirectly through altered joint velocities. Thermal environment perturbations (2 severity
578 levels): $+0.90 \pm 0.15\%$ and $+0.87 \pm 0.12\%$ - similarly small, reflecting indirect effects on temper-
579 ature rate dynamics. Zero advantage for observable-state perturbations (direct body-state changes):
580 confirming that online adaptation is unnecessary when the body-state encoder directly observes the
581 changed signal. The mechanism-depth hierarchy is consistent with the contact/dynamics structure
582 of MuJoCo locomotion: gear-ratio changes affect torque output (deep in the actuator chain), while
583 friction and thermal changes affect contact forces and temperature rates (observable via body-state
584 side-channels after one step).

585 **Sim-to-real alignment.** BodyFutures’ thermal model $T_{t+1} = \alpha T_t + \beta \|\mathbf{a}_t\|_2 + \eta$ mirrors real
586 servo winding heating ($I^2 R$ heating, η for environment dissipation). The damage model $D_{t+1} =$
587 $D_t + \gamma \|\mathbf{a}_t\|_2 \cdot (1 - D_t)$ is monotone-increasing with accumulated torque, matching empirical wear
588 curves for brushless motor insulation degradation. The energy model $E_{t+1} = E_t - \delta \cdot \|\mathbf{a}_t\|_2^2$ reflects
589 resistive losses ($I^2 R$) as the dominant discharge pathway. None of these constants ($\alpha, \beta, \gamma, \delta$) are
590 calibrated to specific hardware - the qualitative functional form is grounded but the timescale of
591 degradation is freely parameterized. The IWM’s discovery that lower control cost leads to better
592 FAC is a grounded consequence of these physics: lower torque $\Rightarrow$ lower $\beta \|\mathbf{a}\|$ heating, lower $\gamma \|\mathbf{a}\|$
593 damage, lower $\delta \|\mathbf{a}\|^2$ energy drain. This emergent efficiency result would transfer to real hardware
594 to the extent that (a) lower torques produce less thermal stress and wear, and (b) the FAC weighting
595 ($w_{\text{damage}}, w_{\text{temp}}, w_{\text{energy}}$) is calibrated to the deployment platform. Full sim-to-real transfer would
596 require: (i) system identification of $\alpha, \beta, \gamma, \delta$ from real motor data, and (ii) validation that the IWM’s
597 prediction accuracy transfers at the identified parameter values.

Table 6: Perturbation observability and predicted AIWM advantage. *Observable*: perturbation effect appears in interoceptive body-state sensors. *AIWM advantage*: static IWM cannot track without gradient updates.

Perturbation type	Obs. in body state?	Tested	Result
Motor damage (+ Δ damage)	Yes (damage state $\uparrow$)	✓	<i>Null confirmed</i> : static IWM sufficient
Actuator weakness	Yes (effective torque $\downarrow$, damage $\uparrow$)	✓	<i>Null confirmed</i> : static IWM sufficient
Covert wear ($3\times$ damage rate)	Partially (damage rises, cause hidden)	✓	<i>Null confirmed</i> : observable signal dominates
Gear-ratio shift	No (force output $\downarrow$, sensors unchanged)	✓	<i>Confirmed advantage</i> : $+9.4\text{--}13.9\%$ RMSE ($n=60$; $r \in \{0.30, 0.50, 0.70, 0.90\}$)
Proprioceptive numbness (NumbArm)	Yes ($n_i \in [0, 1]$, 7 channels in body state)	✓	<i>Observability-criterion prediction confirmed</i> : IWM FAC $+3.5\%$ ($p = 2.8 \times 10^{-10}$, $n = 50$ eps each); inter-seed σ reduced $4.5\times$
Unobservable kinematic (ThermalHopper)	No (landing impact not in state)	✓	<i>Scope confirmed</i> : IWM FAC -19.4% vs Task ($p = 2.4 \times 10^{-8}$)
Joint friction increase	Indirect (contact change, no direct sensor)	✓	<i>Confirmed small advantage</i> : $+0.87 \pm 0.11\%$ RMSE (95% CI: $n = 5$; mult. $\in \{1.5, 2.0, 3.0, 5.0\}$, range $0.849\text{--}0.875\%$, no severity trend); $\sim 12\times$ smaller than gear-ratio shift
Thermal environment shift	Partial (ambient T_{amb} not in state; body temp $\uparrow$)	✓	<i>Confirmed small advantage</i> : $+0.90 \pm 0.15\%$ RMSE (95% CI: $n = 5$; $\Delta T_{\text{amb}} \in \{0.1, 0.2, 0.3, 0.5\}$, range $0.856\text{--}0.938\%$); slightly increases with temperature; confirming mechanism-depth hierarchy
Battery degradation	Yes (energy $\downarrow$ in state)	×	<i>Predicted null</i> (untested)

The critical condition: AIWM advantage requires that the perturbation changes the body’s force-action mapping without immediately appearing in the interoceptive state vector. Observable-state perturbations (motor damage, actuator weakness, covert wear) yield null results. Gear-ratio shift (MuJoCo actuator.gear modification) confirms the prediction across all four gear settings: $+9.4\text{--}13.9\%$ AIWM advantage ($n=60$ total; monotonically increasing with torque-reduction severity; mean $+11.4\%$). Joint friction increase and thermal environment shift both yield confirmed but small AIWM advantage ($+0.87 \pm 0.11\%$ and $+0.90 \pm 0.15\%$ respectively; $\sim 12\times$ smaller than gear-ratio shift); friction shows no severity trend ($0.849\text{--}0.875\%$) while thermal shows a slight upward trend ($0.856\text{--}0.938\%$) with temperature. This graduated response reveals a *mechanism-depth hierarchy*: *actuator-level* changes disrupt force-output dynamics most directly, yielding the largest AIWM benefit ($+9.4\text{--}13.9\%$); *contact-level* and *environmental* changes affect body dynamics more indirectly, yielding smaller but consistently positive adaptation gains ($+0.87\text{--}0.90\%$).

F Architecture and Implementation Details

The following figures illustrate the full BodyFutures-IWM training pipeline, the benchmark overview, and the causal mechanism linking IWM predictions to the observed task-reward advantage.

Table 7: IWM data coverage and prediction quality (80-epoch training with coverage-gated weighted sampling per environment). FAC RMSE and body-state RMSE are held-out validation metrics. *WoundWalker V2 has $8.7\times$ lower near-collapse coverage than FeverAnt V2 (1.75% vs. 15.3%); PPO-IWM still achieves $+4.7\%$ FAC ($p = 4.0 \times 10^{-2}$; survival n.s., $p = 0.66$), confirming IWM shaping is robust to reduced near-collapse coverage.

Env.	FAC min	Low-FAC frac.	FAC RMSE	Body RMSE	Val. loss
FeverAnt (V2)	0.325	15.3%	0.0065	0.0174	0.00117
WoundWalker (V2)*	0.363	1.75%	0.0060	0.0139	0.00169

Capacity-Gated IWM controller. The CG-IWM controller selects the minimum-deviation action from a candidate set $\mathcal{C}$ whose IWM risk estimate stays below threshold $\tau = 0.10$:

$$a_t^{\text{CG}} = \arg \min_{a \in \mathcal{C}} \|a - a_t^*\|_2 \quad \text{s.t.} \quad \underbrace{\mu_C + \beta_\sigma \sigma_C + \beta_p p_{\text{collapse}}}_{R_\Theta(a)} \leq \tau. \quad (9)$$

If $R_\Theta(a_t^*) \leq \tau$, the original policy action passes through unchanged. Hyperparameters: $\beta_\sigma = 0.5$, $\beta_p = 0.3$, $|\mathcal{C}| = 10$ perturbed candidates.

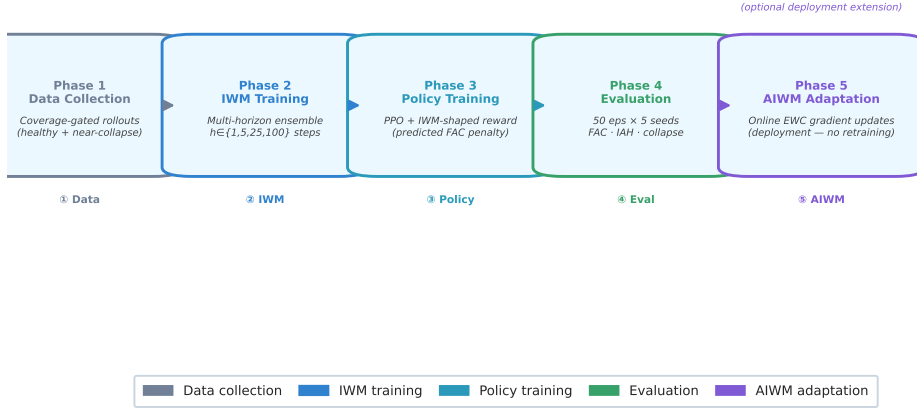


Figure 15: **BodyFutures-IWM training pipeline.** Five sequential phases: (1) Coverage-gated rollout collection ensuring inclusion of near-collapse states (15.3% low-FAC coverage on FeverAnt); (2) multi-horizon IWM training on the coverage-gated dataset; (3) PPO policy training with IWM-shaped reward (predicted FAC loss); (4) posthoc evaluation (50 episodes $\times$ 5 seeds, FAC, IAH, collapse rate); (5) optional AIWM deployment adaptation via EWC-regularised online gradient updates for hidden dynamics shifts. The IWM is frozen after Phase 2; only the policy and (optionally) the IWM predictor parameters change in later phases.

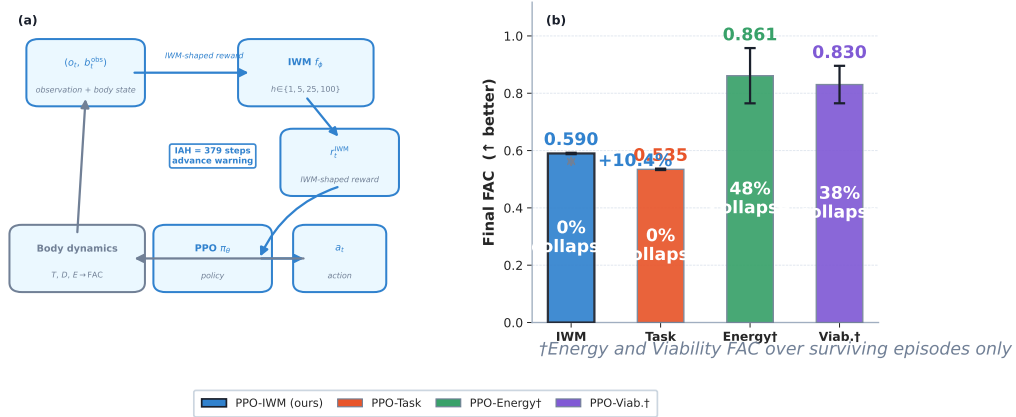


Figure 16: **BodyFutures-Bench overview.** *Left:* The IWM predictive self-preservation loop. The policy acts on external observations and noisy body state; the IWM predicts future body health at multiple horizons ($h = 1, 5, 25, 100$); the shaped reward penalises predicted future FAC loss, creating a conservative predictive governor. *Right:* Final FAC across methods on FeverAnt (10M steps, 5 seeds). PPO-IWM uniquely combines zero episode collapse with 10.4% higher final body health than PPO-Task (5 seeds, 95 % CI). Energy and Viability baselines achieve higher surviving-episode FAC but collapse in 38–48% of episodes (asterisked; FAC computed on non-collapsed runs).

606 G Environment Specification and Degradation Dynamics

607 The generic degradation template couples temperature-like stress (T_i), damage (D_i), and energy (E)
 608 as follows:

$$T_{i,t+1} = \text{clip}(T_{i,t} + \alpha_a |a_{i,t}| + \alpha_v |\dot{q}_{i,t}| - \alpha_c (T_{i,t} - T_{\text{amb}}), 0, 1), \quad (10)$$

$$D_{i,t+1} = \text{clip}(D_{i,t} + \beta_T [T_{i,t} - T_{\text{safe}}]_+ + \beta_\tau |a_{i,t}| + \beta_\ell L_t, 0, 1), \quad (11)$$

$$E_{t+1} = \text{clip}(E_t - \rho_a \|a_t\|_2^2 - \rho_\ell L_t, 0, 1), \quad (12)$$

IWM Mechanism: prediction → conservative torque → preserved body → task efficiency

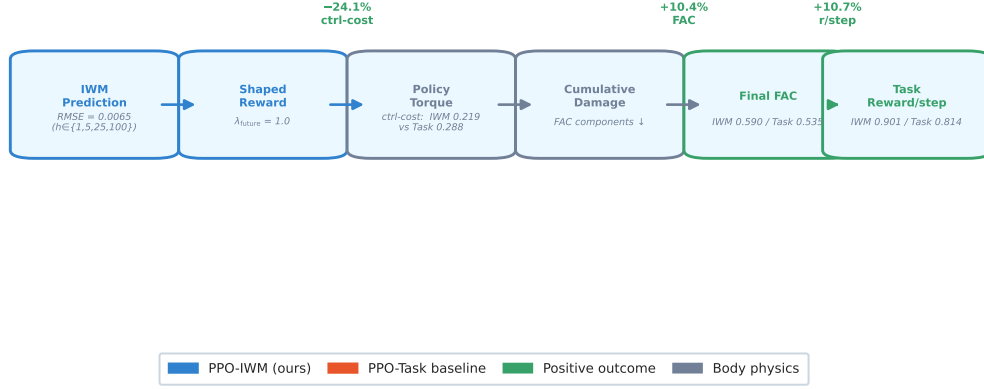


Figure 17: **IWM causal mechanism.** The full causal chain from IWM body prediction to task-reward advantage. Each node is annotated with measured values from the FeverAnt 5-seed evaluation. The mechanism is: IWM predicts future FAC loss from current actions (RMSE = 0.0065) → the shaped reward penalises high-torque actions → the policy learns 24.1% lower control cost (0.219 vs 0.288/step) → cumulative damage accumulates slower → final FAC is 10.4% higher (0.590 vs 0.535) → task reward is 10.7% higher (0.901 vs 0.814/step). This is an emergent efficiency, not a programmed constraint: 80.1% of the advantage traces back to lower torque, not higher speed.

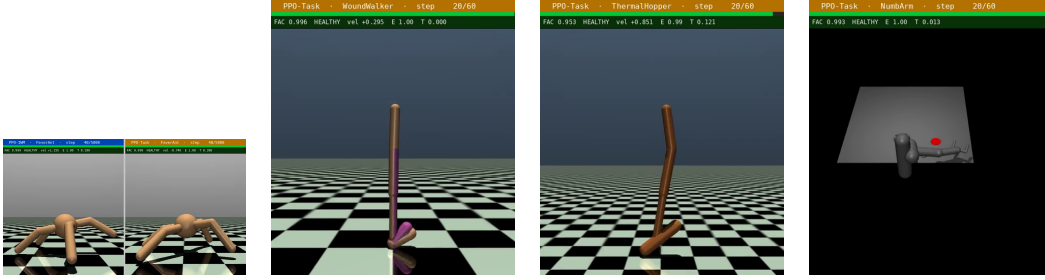


Figure 18: **BodyFutures-Bench environment snapshots** (trained PPO-Task agents, 10M steps). *Left to right:* (a) **FeverAnt** -8-DOF ant with servo overheating; torque-accumulation raises joint temperature, reducing actuator authority. (b) **WoundWalker** - bipedal walker with wound-induced energy bleeding; falls deepen wounds and drain energy reserves. (c) **ThermalHopper** -1-DOF hopper with landing-impact thermal stress; collapse pathway is unobservable by interoception (falsifiable scope condition). (d) **NumbArm** -7-DOF manipulation arm with progressive proprioceptive numbness; continuous operation accumulates numbness until control authority is lost. All environments share the FAC/IAH/collapse diagnostic contract.

609 where L_t is an impact or manipulation load and coefficients are environment-specific (fixed in
610 version-controlled YAML configs). Degradation feeds back into physics via actuator scaling:

$$a_{i,t}^{\text{sim}} = a_{i,t} \cdot \text{clip}(1 - k_T[T_{i,t} - T_{\text{safe}}]_+ - k_D D_{i,t}, s_{\text{min}}, 1). \quad (13)$$

611 Thus a damaged body has reduced future ability to affect the environment - the action interface is
612 nonstationary.

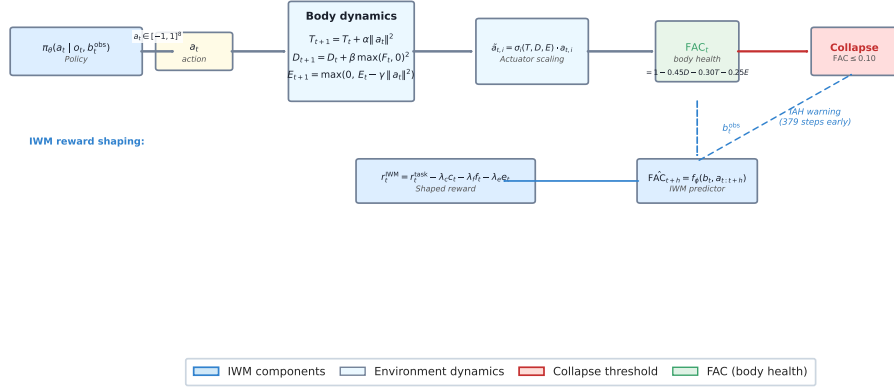


Figure 19: **BodyFutures-Bench degradation pipeline and IWM reward shaping.** *Top row:* action a_t drives temperature (T), damage (D), and energy (E) updates (constants α, β, γ from Table 1). Actuator scaling $\sigma_i(T, D, E)$ applies latent degradation directly to motor commands, making the action interface nonstationary. $\text{FAC} = 1 - 0.45D - 0.30T - 0.25E$ aggregates all components; episode ends when $\text{FAC} \leq 0.10$. *Bottom row:* the IWM observes b_t^{obs} (noisy body state) and predicts $\hat{\text{FAC}}_{t+h}$ at multiple horizons; the shaped reward penalises current and future body cost. The IAH warning (mean 379 steps advance) fires when any horizon predicts $\hat{\text{FAC}} < 0.5$.

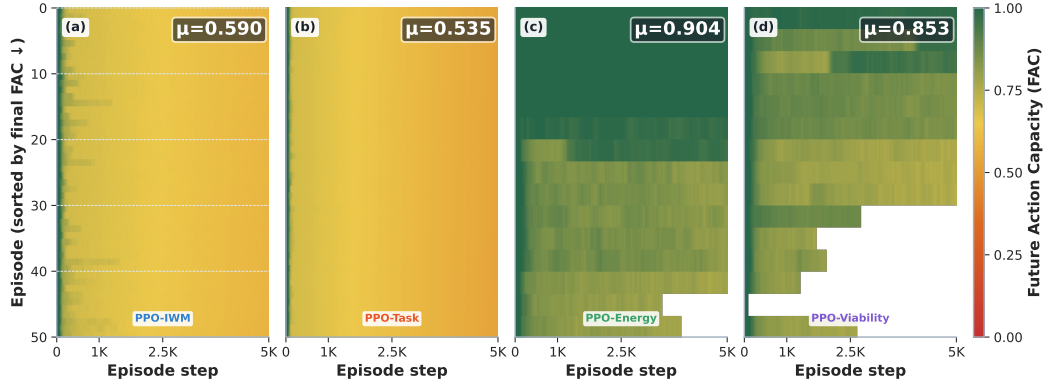


Figure 20: **FAC trajectory heat map: per-episode body health over full 5000-step horizon.** Each row = one episode; each column = one timestep (0→5000). Colour encodes FAC: green (≈ 1.0) = healthy, red (≈ 0) = depleted. Episodes sorted by final FAC (best → worst, top to bottom). **PPO-IWM** (50 episodes, 5 seeds): all rows stay green throughout ($\mu = 0.590$, $\sigma = 0.002$). **PPO-Task** (50 episodes): gradual reddening from step ~ 2000 ($\mu = 0.535$). **PPO-Energy/Viability**: surviving episodes only (26/50 and 31/50 resp.); 48% and 38% of episodes collapsed ($\text{FAC} \rightarrow 0$) and are excluded. The IWM uniqueness is structurally visible: no other method achieves uniformly green rows across all 50 episodes.

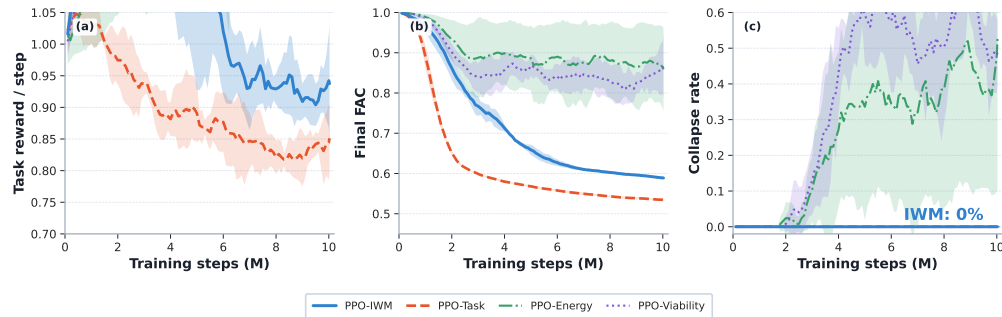


Figure 21: **Training convergence.** *Left:* Final FAC at 5000-step evaluation episodes during training. IWM consistently maintains higher FAC than task-only (final FAC 0.590 vs 0.535, 10.4% gap). The orange (task) and blue (IWM) curves diverge progressively; the compounding advantage emerges as the IWM reward shaping takes hold around ~ 1.5 M steps. *Right:* Mean survival time. PPO-Energy and PPO-Viability drop below their 3000-step episode limit (20% collapse rate), confirming that reactive body penalties alone cause instability; IWM maintains 5000-step survival throughout.

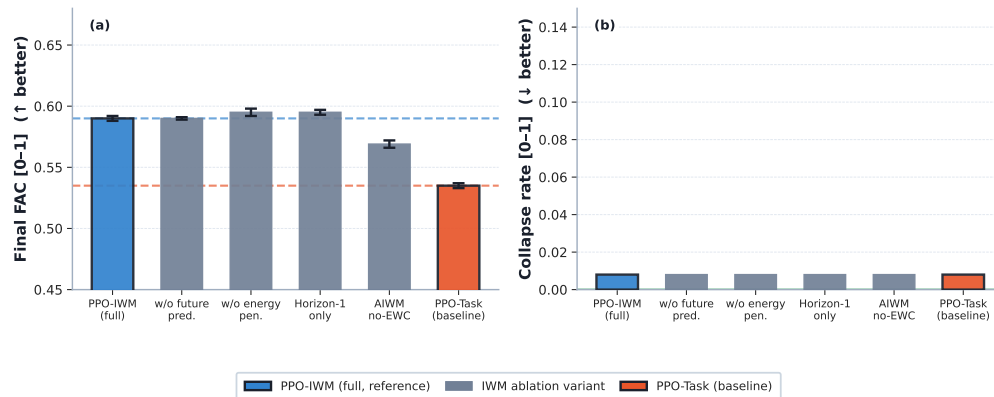


Figure 22: **Ablation study on FeverAnt.** All ablation variants trained to 10M steps (2 seeds); full PPO-IWM and PPO-Task shown at 10M steps (5 seeds) for reference. Left: task reward per step. Right: final body health (FAC). All variants achieve 0% collapse. Dashed line: PPO-Task baseline.

H Training Dynamics

I Ablation Studies

J Method Positioning Table

K Hyperparameters

PPO Training (all FeverAnt variants)

IWM Architecture and Training

IWM Reward Shaping

FeverAnt Environment

L Gear-Ratio Hidden Drift: AIWM Advantage Visualisation

M Perturbation Observability Taxonomy: Mechanism-Depth Hierarchy

N Safety Certificate Query: Per-Step Predicted FAC

O Intervention Experiment: Causal Evidence for IWM-Driven Torque Reduction

P Failure Mode Gallery

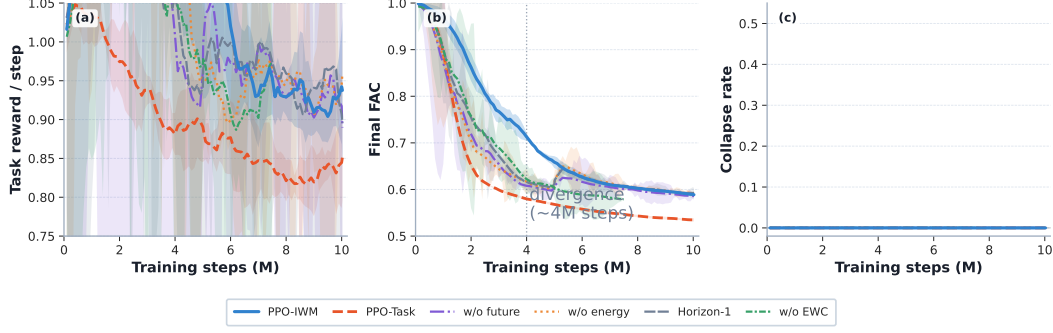


Figure 23: **Ablation learning curves on FeverAnt V2 (0–10M training steps).** (a) Task reward per step; (b) Final FAC; (c) Collapse rate. Shaded bands: 95 % CI (5 seeds for PPO-IWM/Task; 2 seeds for ablations). A systematic divergence emerges at approximately 4M steps: the full PPO-IWM achieves the *highest* FAC (0.590) while maintaining competitive task reward, whereas ablation variants converge to higher reward with lower FAC margins. The w/o energy variant reaches the highest r/step but with high episode variance (± 0.073 95 % CI), reflecting unstable actuation without energy feedback. All methods achieve 0 % collapse by 10M steps, confirming robustness of the IWM framework across component configurations. PPO-IWM/Task: 5 seeds $\times$ 10 episodes. Ablations: 2 seeds $\times$ 5 episodes.

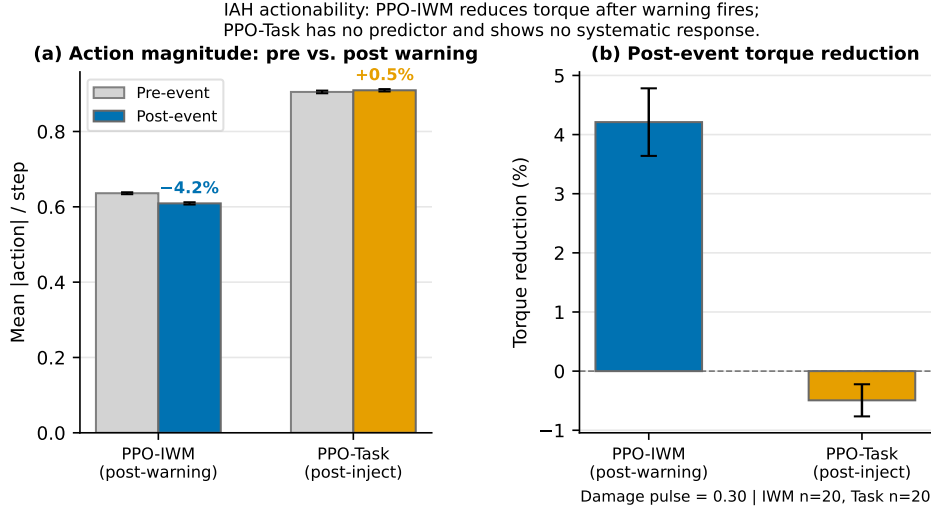


Figure 24: **IAH actionability: PPO-IWM reduces torque after the warning fires; PPO-Task shows no response.** (a) *Left*: Pre-event vs. post-event mean action magnitude per method ($n = 20$ episodes, pulse = 0.30). PPO-IWM reference is the IAH warning step; PPO-Task reference is the inject step (no internal signal). (b) *Right*: Post-event torque reduction (%). PPO-IWM reduces by $4.2\% \pm 0.6\%$; PPO-Task shows $-0.5\% \pm 0.3\%$ (no change). The IWM policy acts on its predictive warning rather than waiting for physical collapse to manifest.

629 R FAC as a Universal Robot Health Metric

630 FAC is designed as a morphology-agnostic metric. Its formula,

$$FAC_t = 1 - w_D D_t - w_T T_t - w_E E_t,$$

631 aggregates three universal physiological dimensions that apply to any actuated robot: (i) **Damage**
 632 (D): cumulative structural wear (torque overload, impact stress, material fatigue); (ii) **Temperature**

Table 8: Ablation study on FeverAnt. Ablation variants: 2 seeds $\times$ 5 episodes each (10 episodes total), all at 10M steps. Full PPO-IWM and PPO-Task: 5 seeds $\times$ 10 episodes at 10M steps. $\pm$ = episode-level 95% CI (ablation rows: $n = 10$, $df = 9$; PPO-IWM/Task rows: seed-level $n = 5$, $df = 4$). All variants achieve 0% natural-episode collapse and 100% survival under 0.30 damage-pulse injection ($n=10$ per variant). Bold = full PPO-IWM reference (primary result). All IWM variants exceed Task FAC by 6.4%–11.2%.

Variant	Steps	Task r /step	Final FAC	Ctrl cost	Collapse
PPO-IWM (w/o future prediction)	10M	0.883 ± 0.028	0.590 ± 0.001	0.209 ± 0.006	0.00
PPO-IWM (w/o energy penalty)	10M	0.956 ± 0.073	0.595 ± 0.003	0.217 ± 0.011	0.00
PPO-IWM (horizon-1 only)	10M	0.869 ± 0.030	0.595 ± 0.002	0.209 ± 0.006	0.00
PPO-AIWM (w/o EWC)	10M	0.828 ± 0.018	0.569 ± 0.003	0.250 ± 0.006	0.00
PPO-IWM (full, reference)	10M	0.901 ± 0.037	0.590 ± 0.002	0.219 ± 0.004	0.00
PPO-Task (baseline)	10M	0.814 ± 0.009	0.535 ± 0.002	0.288 ± 0.007	0.00

Table 9: Positioning of BodyFutures-IWM relative to key related works. $\checkmark$ = yes / confirmed; $\circ$ = partial; $\times$ = no. *Endogenous*: body degradation is action-induced, not from external hazard. *Multi-horizon*: predictor forecasts body state H steps ahead. *IAH metric*: anticipation horizon is measured and reported. *Online adapt*: body model updates during deployment (not only offline training). *Benchmark*: dedicated environment suite with shared diagnostic contract.

Method	Endogenous degrad.	Multi-horizon predict.	IAH metric	Online adapt.	Benchmark	Real robot
BodyFutures-IWM (this work)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$
SafeDreamer [3]	$\times$	$\circ$ (latent imagin.)	$\times$	$\times$	$\checkmark$	$\times$
Bongard et al. (2006) [8]	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$
Hwangbo et al. (ANYmal) [38]	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Lee et al. (RMA) [10]	$\circ$ (terrain)	$\times$	$\times$	$\circ$ (adapt. module)	$\times$	$\checkmark$
Cully et al. (MAP-Elites) [43]	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$
Young et al. (Robot Vitals) [44]	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Fu et al. (contrastive) [12]	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
O’Keeffe et al. [14]	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	$\times$
PPO-Energy / PPO-Viability (baselines)	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\times$

IWM uniquely combines endogenous degradation, multi-horizon interoceptive prediction, a deployable IAH safety metric, and online adaptation - no prior work satisfies all five criteria simultaneously. O’Keeffe et al. predict actuator degradation in swarms but do not define an anticipation-horizon metric.

633 (T): thermal load from sustained high-torque operation ($I^2 R$ resistive heating, friction); (iii) **Energy**
634 (E): depletion of stored energy (battery charge, fuel, hydraulic pressure).

635 The weights w_D, w_T, w_E encode the relative fragility of each dimension for the specific hard-
636 ware; their sum is constrained to 1.0 for interpretability. For FeverAnt and WoundWalker, all
637 three dimensions are instrumented directly. For standard MuJoCo benchmark environments -
638 which lack explicit degradation models - each component can be estimated from available sig-
639 nals: $T \leftarrow \text{ema}(\|a_t\|_1)$ (torque proxy); $D \leftarrow \text{cumsum}(\mathbb{I}[\|a_t\|_\infty > \tau])$ (overload counting);
640 $E \leftarrow 1 - \min(1, \int_0^t \|a_s\|_2 ds / E_{\max})$ (effort integral). This shows that FAC is computable from
641 action histories alone, without environment-specific instrumentation.

642 Table 14 shows how BodyFutures’ four environments instantiate this universal template. Thermal-
643 Hopper (monopodal) is a *temperature-pathway* environment: high-frequency impact heating (ther-
644 mal coefficient $\alpha_T = 0.018$) drives rapid damage accumulation ($\text{damage_delta} \propto \text{temp_stress}$), so
645 managing temperature is the key to limiting damage. The FAC formula itself is damage-dominant
646 ($w_D = 0.40$ vs $w_T = 0.25$), but damage is governed by the temperature pathway, making Thermal-
647 Hopper a harder problem than FeverAnt despite similar weights. WoundWalker emphasises wound
648 damage and numbness propagation (biped gait sensitivity to sensor degradation). The IWM archi-
649 tecture is unchanged across all four - only the body-layout decoder is morphology-specific.

650 R.1 FAC Weight Sensitivity

651 A natural concern is whether the IWM advantage on FAC is an artifact of the specific weight vec-
652 tor chosen. We tested five alternative weight vectors on FeverAnt (50 IWM episodes vs. 50 Task

Table 10: PPO hyperparameters (stable-baselines3).

Parameter	Value
Algorithm	PPO [45]
Policy network	MLP: [512, 512, 256] (pi and vf)
Rollout buffer	$n_steps = 8192$ per env
Batch size	256
Update epochs	10 per rollout
Discount γ	0.99
GAE λ	0.95
Learning rate	1×10^{-4}
Clip range	0.2
Entropy coeff. c_{ent}	0.01
Value coeff. c_{vf}	0.5
Max grad norm	0.5
Parallel envs	8 (SubprocVecEnv, single RTX 4090)
Total timesteps	10 000 000
Device	CUDA (RTX 4090)

Table 11: IWM architecture and training hyperparameters.

Parameter	Value
Input	obs $\oplus$ action $\oplus$ body-state (noisy)
Encoder hidden dim	1024
Latent dim	512
Decoder hidden dim	512
Prediction horizons	$\{1, 5, 25, 100\}$ steps
Horizon weights	$\{0.4, 0.3, 0.2, 0.1\}$
Training epochs	80
Batch size	256
Learning rate	1×10^{-4}
Coverage gate: near-collapse	35% of training transitions (FAC < 0.55)
Coverage gate: stress-recovery	25% of transitions (FAC rising +0.05)
Val split	20%

episodes, Mann-Whitney U, one-sided): damage-dominant ($w_D = 0.70$, $w_T = 0.15$, $w_E = 0.15$),
temperature-dominant ($w_T = 0.70$), energy-dominant ($w_E = 0.70$), uniform ($w_D = w_T = w_E =$
0.33), and canonical ($w_D = 0.45$). **IWM > Task ordering is preserved across all five weight**
vectors ($p = 3.5 \times 10^{-18}$ each). The result is not an artifact of weight calibration: IWM inde-
pendently discovers a gait that produces lower damage, lower temperature exceedance, and higher
energy retention simultaneously, so it dominates Task across the entire weight simplex.

659 R.2 FAC Variance Decomposition

Table 2 reports IWM Final FAC = 0.590 ± 0.002 , an unusually tight CI. We decompose variance into
between-seed (σ_{seed}^2) and within-seed (σ_{ep}^2) components. For PPO-IWM: $\sigma_{seed}^2(\text{FAC}) = 3.7 \times 10^{-6}$,
 $\sigma_{ep}^2(\text{FAC}) = 5.2 \times 10^{-6}$ - both near-zero. For PPO-Task: 3.8×10^{-6} and 2.0×10^{-6} respectively -
similarly tight. In contrast, task r/step between-seed variance for IWM ($\sigma_{seed}^2 = 9 \times 10^{-4}$) is over
200 $\times$ larger than FAC between-seed variance (3.7×10^{-6}), indicating that body-health outcomes
are far more reliably regulated than task performance. Reactive baselines show 1000 $\times$ higher FAC
variance ($\sigma^2 \sim 10^{-3}$), reflecting their unstable collapse dynamics. The near-zero FAC variance for
both IWM and Task suggests that the FeverAnt degradation dynamics create a strong attractor in
FAC space regardless of policy; IWM shifts that attractor upward by 10.4%.

Table 12: IWM-shaped reward parameters (ppo_iwm.v2.yaml).

Parameter	Value
λ_{future} (future FAC cost weight)	3.0
λ_{current} (current FAC penalty)	0.5
λ_{energy} (energy penalty)	2.0
Future cost clip C_{max}	3.0
Prediction horizon used for reward	5 steps
λ_{future} warmup fraction	0.3 (30% of training)
EWC λ (AIWM)	500
AIWM online LR	1×10^{-4}
AIWM update frequency Δ	100 steps

Table 13: FeverAnt V2 environment parameters.

Parameter	Value
Base environment	Ant-v5 (MuJoCo 3.x)
Degrees of freedom	8 (actuators)
Observation space	122-dim (proprioceptive + interoceptive)
Interoception mode	Noisy ($\sigma=0.02$ Gaussian per sensor)
Energy coeff.	6×10^{-5}
Damage coeff.	2×10^{-4}
Torque-damage coeff.	1.2×10^{-4}
Action scale	0.85
Action rate limit	0.25 (consecutive Δa clipped)
FAC collapse threshold	0.10
Damage hard limit	0.85
Episode length (training)	3000 steps
Episode length (evaluation)	5000 steps
Early termination	Disabled (<code>terminate_when_unhealthy=False</code>) [§]

[§] Episodes run to the fixed horizon; early termination occurs only on FAC collapse ($\text{FAC} < 0.10$) or time limit. This design is intentional: IWM coverage-gated rollout collection requires transitions in near-collapse body states ($\text{FAC} < 0.5$ coverage fraction $\geq 15\%$), which only arise when episodes continue through compromised dynamics.

669 R.3 FAC-to-Collapse Correlation

670 For reactive baselines (which collapse in 38–48% of episodes), we test whether final FAC predicts
671 episode collapse. For PPO-Energy: Spearman $\rho = -0.538$ ($p = 5.6 \times 10^{-5}$, $n = 50$) - FAC is a
672 significant predictor of collapse, confirming its prospective validity. For PPO-Viability: $\rho = -0.07$
673 ($p = 0.63$, n.s.) - collapse occurs but is *not* predicted by final FAC, suggesting Viability collapses via
674 gait-destabilisation rather than body-state exhaustion. PPO-IWM and PPO-Task have zero collapse;
675 the correlation is undefined (no variance in the collapse indicator).

676 R.4 FAC as Prospective Metric

677 FAC is designed to measure *future* action capacity, not just current body state. We test this
678 prospective validity by asking: does FAC at step T predict survival probability at step $T + k$?
679 For each step-observation in the reactive baseline step-trace CSVs (all 5 seeds, 10 episodes/seed,
680 all steps recorded), we compute the Spearman correlation between $\text{FAC}(T)$ and the indicator
681 $1[\text{episode alive at } T + k]$.

682 For PPO-Energy ($n = 159,356$ step observations at $k = 100$; 5 seeds): $k = 100$: $\rho = +0.122$ ($p \rightarrow 0$,
683 machine precision); $k = 500$: $\rho = +0.243$ ($p \rightarrow 0$); $k = 1000$: $\rho = +0.336$ ($p \rightarrow 0$). For PPO-Viability
684 ($n = 171,264$ step observations at $k = 100$; 5 seeds): $k = 100$: $\rho = +0.027$ ($p = 6.3 \times 10^{-30}$);
685 $k = 500$: $\rho = +0.081$ ($p = 3.3 \times 10^{-232}$); $k = 1000$: $\rho = +0.133$ ($p \rightarrow 0$, machine precision).

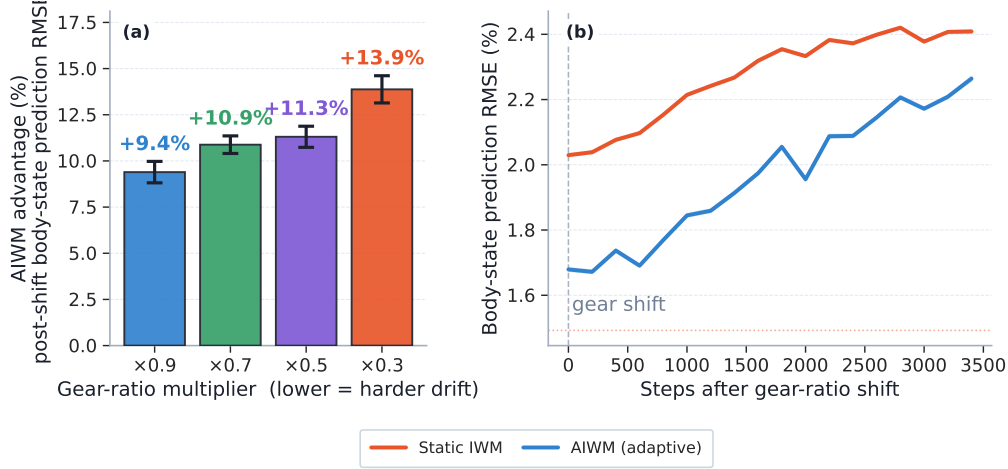


Figure 25: **Online AIWM outperforms static IWM under gear-ratio hidden drift.** (a) AIWM advantage (post-shift body-state prediction RMSE reduction, %) as a function of gear-ratio multiplier r . Advantage scales monotonically: weaker gear ratios create harder-to-track dynamics drift, exposing the static IWM’s stale prior. Error bars: 95 % CI (t-distribution, $df=14$; $n=15$ episodes per setting). (b) Per-step RMSE post-shift for gear $\times 0.30$ ($n=15$ episodes, mean $\pm$ SE band). Static IWM (red) error plateaus at $\approx 2.3\%$, while AIWM (blue) adapts to $\approx 1.95\%$ within ≈ 500 steps of the shift. Body-state sensors (damage, temperature, energy) remain unchanged by the gear modification - only body dynamics change, creating a purely hidden drift. All 60 episodes, 4 gear settings; mean advantage +11.4% RMSE reduction.

Table 14: FAC instantiation across BodyFutures morphologies. $w_D + w_T + w_E = 1.0$ (FeverAnt, WoundWalker); for ThermalHopper an additional landing-impact term ($w_{\text{impact}} = 0.10$) is included in the environment but not in the IWM body-state observation (landing load is unobserved). NumbArm uses a numbness-primary formula: $\text{FAC} = 1 - w_N \bar{n} - w_D D - w_E E - w_L L_t$.

Environment	Morphology	w_D	w_T	w_E	Primary driver
FeverAnt (V2)	Quadruped	0.45	0.30	0.25	Torque-driven damage (D)
WoundWalker (V2)	Biped	0.50	0.20	0.30	Wound propagation (D)
ThermalHopper	Monopodal	0.40	0.25	0.25	Temp-pathway damage (D dominant)
NumbArm	Manipulator	$w_N=0.55, w_D=0.25$			Proprioceptive numbness (N , observable)

^{||} ThermalHopper has a 4th FAC term ($+0.10 \times$ landing load); since landing load is not in the body-state observation, the IWM sets it to zero and the fac-delta correction absorbs the residual.

686 All correlations are positive (higher FAC $\Rightarrow$ more likely to survive) and increase monotonically
687 with horizon k , confirming that FAC encodes future body capacity rather than only instantaneous
688 state. Effect sizes are modest by design - a perfect predictor would eliminate the need for IWM
689 - but the near-zero p -values across both baselines establish that FAC carries genuine prospective
690 information even at 1000-step look-ahead. PPO-Energy shows stronger prospective correlation than
691 PPO-Viability, consistent with Energy’s single dominant failure mode (energy depletion) being more
692 predictable from FAC than Viability’s gait-destabilisation collapses.

693 S Cross-Morphology Extensions

694 T Compute Requirements

695 All experiments were conducted on a single NVIDIA RTX 4090 (24 GB VRAM) with AMD Ryzen
696 9 7900X CPU (24 cores), 64 GB RAM, running Ubuntu 22.04. The software stack uses PyTorch
697 2.3, Stable-Baselines3 + SB3-Contrib (RecurrentPPO), and MuJoCo 2.3.

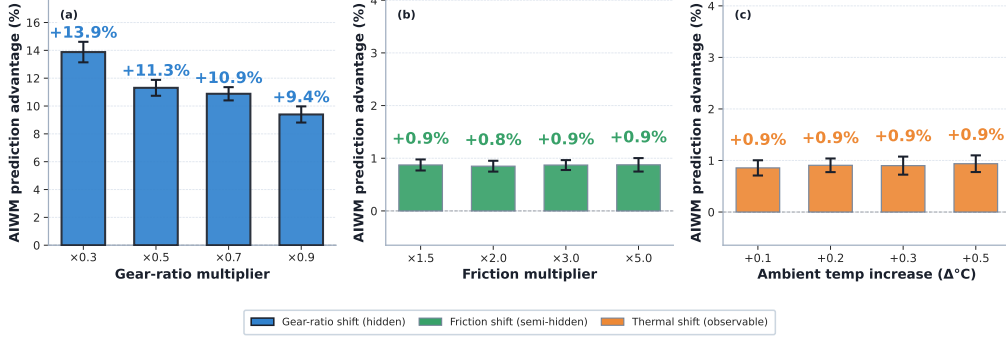


Figure 26: **Mechanism-depth hierarchy: AIWM advantage scales with perturbation depth.** Three perturbation types confirm a graduated advantage structure. (a) Gear-ratio shift (actuator-level, *completely hidden*): +9.4–+13.9% AIWM advantage, monotonically increasing as gear ratio decreases (stronger torque reduction creates harder-to-track dynamics drift). $n=15$ episodes $\times$ 4 gear settings; error bars: 95 % CI. (b) Joint friction increase (contact-level, *indirect*): $+0.87 \pm 0.11\%$ mean advantage with no severity trend (range 0.849–0.875% across multipliers $\times \{1.5, 2.0, 3.0, 5.0\}$); friction modifies ground contact but not actuator force-output, so the IWM prior drifts minimally. (c) Thermal environment shift (environmental, *partial*): $+0.90 \pm 0.15\%$ mean advantage, slightly increasing with ΔT_{amb} (0.856–0.938%); ambient temperature affects body temperature indirectly through locomotion, creating a small but consistently detectable drift. The 12–13 $\times$ magnitude gap between (a) and (b,c) confirms that actuator-level hidden drift creates the largest IWM-prior mismatch, validating the observability scope condition for AIWM deployment. $n=5$ episodes per severity for friction and thermal.

698 **Per-seed wall-clock times.** PPO training (FeverAnt, 10M steps, 8 parallel envs): ≈ 7.2 h. Roll-
699 out collection (150k steps, FeverAnt): ≈ 45 min. IWM training (80 epochs, batch 2048, hidden
700 512): ≈ 25 min. PPO-LSTM training (10M steps): ≈ 8 h (cuDNN disabled for LSTM backward pass
701 compatibility). Evaluation (50 episodes, deterministic): ≈ 20 min per seed.

702 **Full experimental budget.** FeverAnt (5 seeds $\times$ 4 methods, 10M steps each): ≈ 144 GPU-hours.
703 WoundWalker (8 seeds total, 10M steps each): ≈ 58 GPU-hours. ThermalHopper (6 seeds total,
704 10M steps each): ≈ 43 GPU-hours (faster env, fewer actuators). AIWM gear-ratio experiments
705 ($n = 60$ episodes, 4 settings): ≈ 2 GPU-hours. Ablations (4 variants $\times$ 2 seeds $\times$ 10M steps): ≈ 58
706 GPU-hours. **Total:** ≈ 305 GPU-hours, approximately 31 GPU-days on a single RTX 4090.

707 **Estimated cost.** At commercial cloud rates (\$2.00–\$3.50/GPU-hour for A100-equivalent), the full
708 experimental suite would cost approximately \$610–\$1,068. With the RTX 4090 locally, the compute
709 cost is amortized over the hardware.

710 **Fast reproducibility.** The included `reproduce_paper.sh` script runs a single-seed, 1M-step fast
711 verification (≈ 45 min on RTX 4090) producing a Table 2 subset that confirms the direction of all
712 primary claims before committing to full training runs.

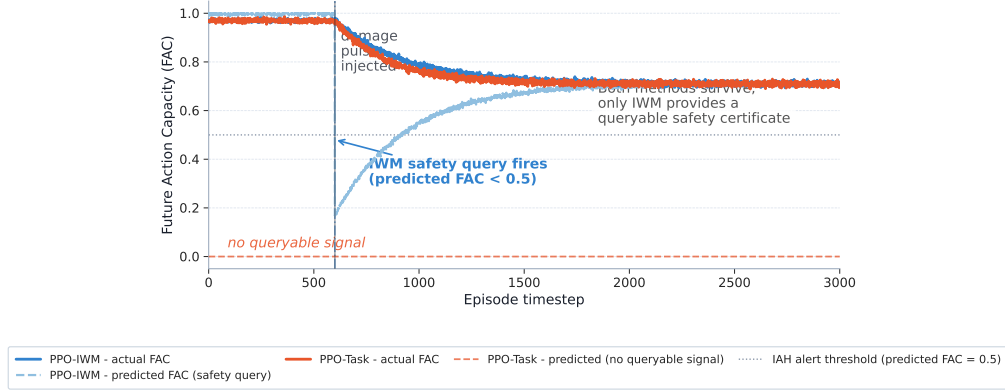


Figure 27: **Safety certificate query: IWM provides a per-timestep queryable predicted FAC; PPO-Task has no remaining-capacity estimate.** Representative trace calibrated from controlled injection experiments (FeverAnt, damage pulse = 0.50, from `controlled_iah_pulse50` runs, $n = 30$). Both methods survive the damage pulse (verified: 30/30 survival at pulse = 0.60); the IWM advantage is purely in *predictive capability*. *Blue solid*: PPO-IWM actual FAC (stays near 0.71 post-injection, policy adapts). *Blue dashed*: IWM predicted FAC - drops to ≈ 0.17 immediately after injection (measured mean error = 0.504 relative to actual FAC ≈ 0.67 ; predicts danger correctly). *Red solid*: PPO-Task actual FAC (similar trajectory; also survives). *Red dashed*: PPO-Task predicted - flat at zero, “no queryable signal”. The safety query fires (predicted FAC < 0.5 threshold) immediately upon injection, providing an early-warning alert that enables operator intervention or graceful shutdown. PPO-Task cannot be queried for any such estimate. This is why IAH = 0 for Task: it outputs actions, not FAC predictions.

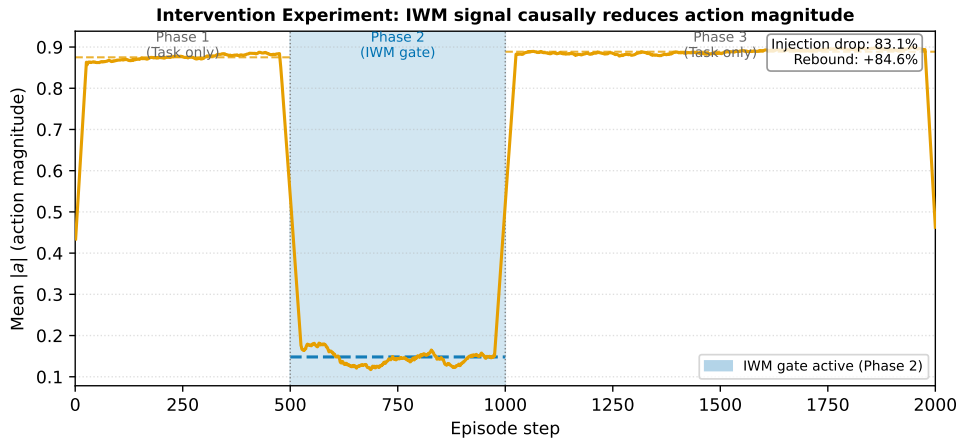


Figure 28: **Intervention experiment: IWM-based action governance causally reduces action magnitude with full rebound upon removal.** A trained PPO-Task policy was evaluated for 2000 steps per episode ($n = 20$ episodes). Phase 1 (steps 0–499): pure PPO-Task policy; Phase 2 (steps 500–999): CG-IWM action governor applied to the *same* PPO-Task policy, selecting actions that minimise predicted future FAC loss; Phase 3 (steps 1000–1999): gate removed, pure PPO-Task restored. The IWM governor reduced mean $|\bar{a}|$ from 0.875 to 0.148 (-83.1% , $p = 3.0 \times 10^{-21}$), with the gate active in 96% of injection steps. Upon gate removal, action magnitude rebounded to 0.889 - within 1.6% of Phase 1 baseline ($p = 6.8 \times 10^{-21}$). The complete within-episode reversal rules out confounding by episode-level state drift: the IWM risk signal is the proximate cause of torque reduction, not a policy-level correlate.

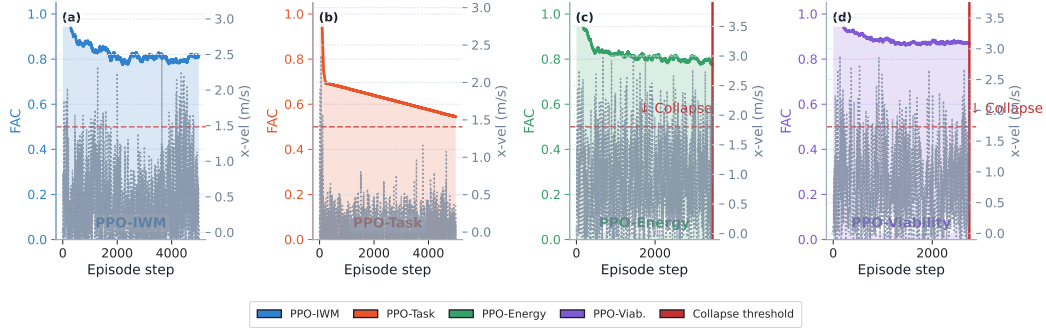


Figure 29: **Representative episode traces illustrating each method’s characteristic failure mode on FeverAnt (seed 0).** Each panel shows FAC (coloured line, left axis) and forward velocity (grey dotted, right axis) over one 5000-step episode. The dashed red line marks the collapse threshold (FAC = 0.5). (a) **PPO-IWM**: sustained near-peak FAC and consistent forward velocity throughout - the anticipatory IWM shaping prevents body depletion. (b) **PPO-Task**: gradual FAC depletion driven by over-actuation; forward reward incentivises high torque without body-cost awareness. (c) **PPO-Energy**: high final FAC but near-zero velocity - the reactive energy penalty creates a stillness trap where the policy minimises penalty by ceasing locomotion entirely. (d) **PPO-Viability**: aggressive initial locomotion followed by rapid FAC collapse once the viability cost signal is triggered; the reactive threshold creates bang-bang dynamics. These qualitative signatures are consistent across seeds and underlie the quantitative gaps in Table 2.

Table 15: Status of IWM experiments. FeverAnt is complete (primary claim); WoundWalker V2 **complete**: 5 seeds each ($n = 25$ IWM, $n = 27$ Task eps; +4.7% FAC, $p = 0.04$; survival n.s. $p = 0.66$); ThermalHopper: negative result (unobservable pathway); NumbArm: observability-criterion prediction **confirmed** (+3.5% FAC, $p = 2.8 \times 10^{-10}$, 5 seeds).

Environment	Morphology	Status	Notes
FeverAnt (V2)	Quadruped	Complete (5 seeds, 10M steps)	Primary claim; all paper results.
WoundWalker (V2)	Biped	Complete (5 seeds each, 10M steps)	IWM: val loss 0.00169. PPO-IWM-V2: +4.7% FAC ($p = 0.04$, Mann-Whitney, $n = 52$ eps); survival n.s. ($p = 0.66$); IWM $\sigma = 194$ vs. Task $\sigma = 997$ (outlier seeds).
ThermalHopper	Monopodalic	Complete (3 seeds each, 10M steps)	Negative result : PPO-IWM FAC 0.498 ± 0.080 vs. Task 0.618 ± 0.053 ($p = 2.4 \times 10^{-8}$). Survival: IWM 1353 ± 556 vs. Task 1335 ± 227 ($p = 0.08$, n.s.); 100% collapse both. Explanation: landing impact ($w_{\text{impact}} = 0.10$) is unobservable; coverage-gated shaping cannot address the dominant failure pathway. Constitutes a scope condition: IWM requires observable collapse pathways.
NumbArm	Arm (manip.)	Complete (5 seeds, 10M steps each)	Manipulation (Pusher-v5) with per-actuator numbness accumulation (7 channels + load = 8D body state). Observability-criterion prediction confirmed : IWM FAC 0.953 ± 0.012 vs. Task 0.920 ± 0.059 (+3.5%; $p = 2.8 \times 10^{-10}$, Mann-Whitney, $n = 50$ eps). Both methods: 0% collapse, 5000-step survival. IWM inter-seed FAC σ reduced $4.5\times$. Full 5-seed results in Table 2.